Homework 8

Nathan Walker

1. This problem explores the design a symmetric real-valued linear phase FIR bandpass filter that meets the following specifications.

filter design	Length	Order	group delay	type	window parameters
Window	62	61	30.5	II	hann
Kaiser	46	45	22.5	II	$\beta = 3.3953$
equiripple	30	29	14.5	II	equiripple

Window Design

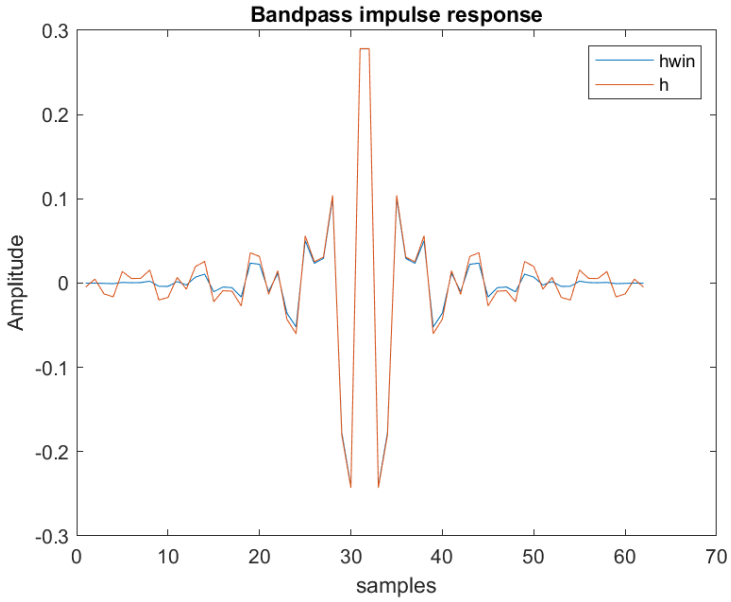


Figure 1: impulse response of the windowed and non windowed filter

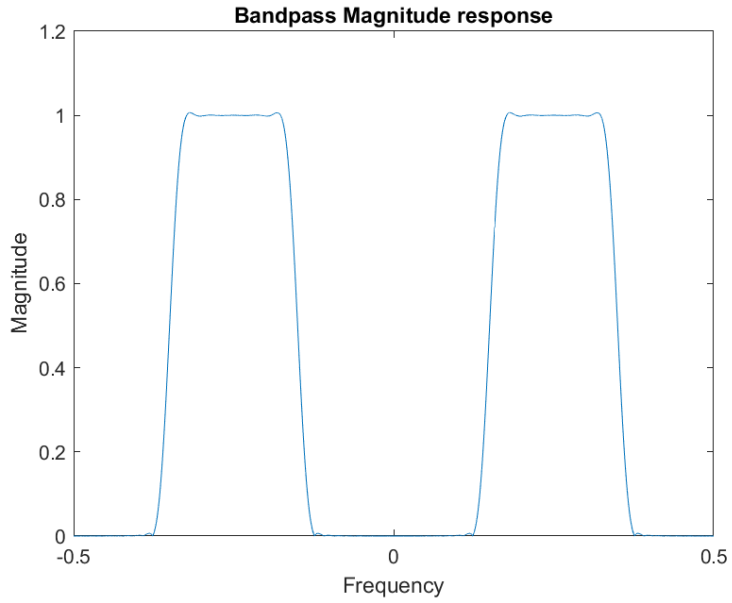


Figure 2: Linear magnitude response of the windowed filter

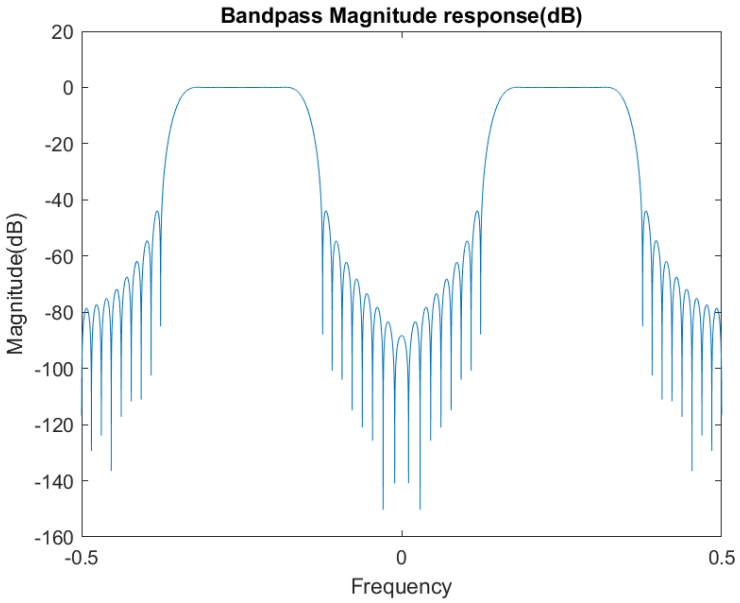


Figure 3: Logrithmic magnitude response of the windowed filter

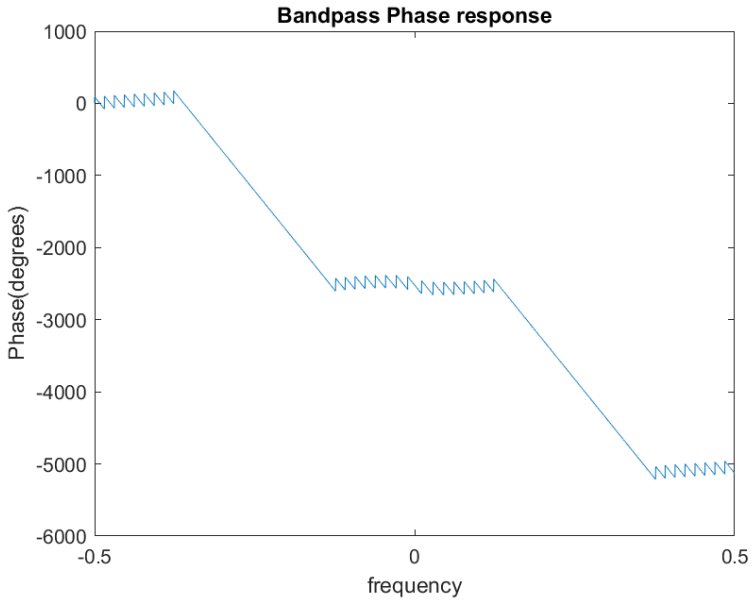


Figure 4: Phase response of the windowed filter

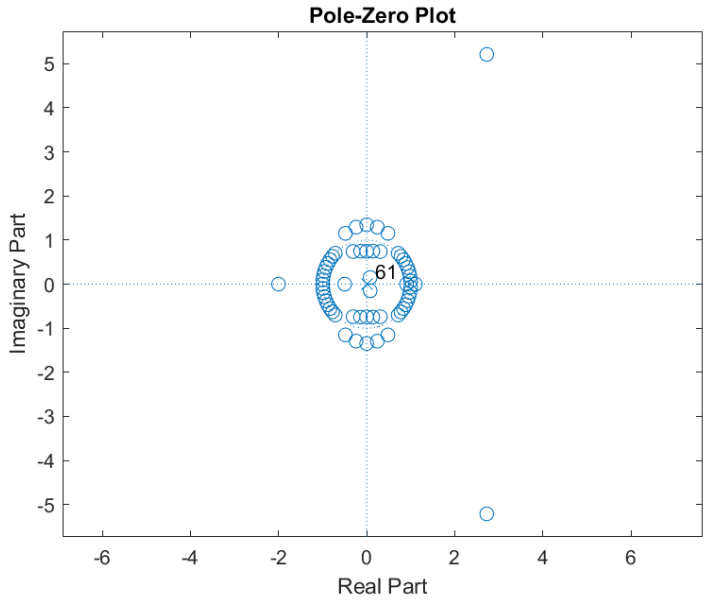


Figure 5: Pole zero plot of the windowed filter

Kaiser

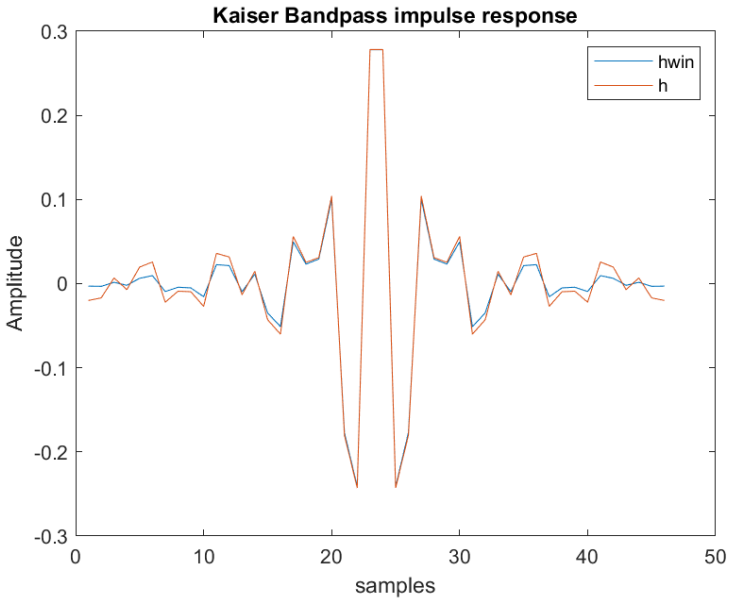


Figure 6: impulse response of the windowed and non windowed filter

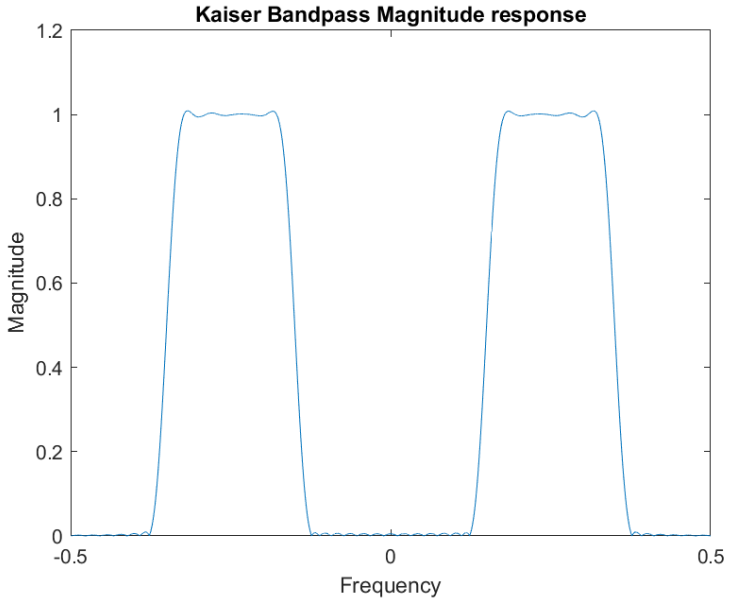


Figure 7: Linear magnitude response of the windowed filter

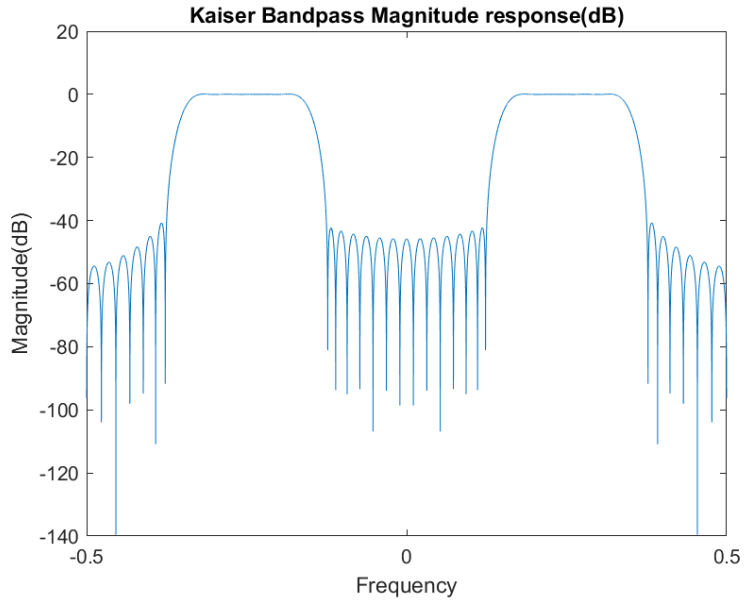


Figure 8: Logrithmic magnitude response of the windowed filter

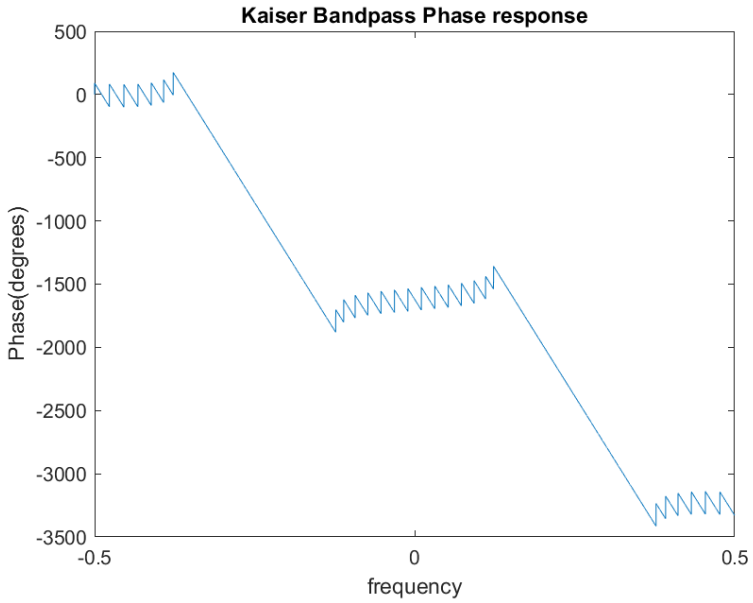


Figure 9: Phase response of the windowed filter

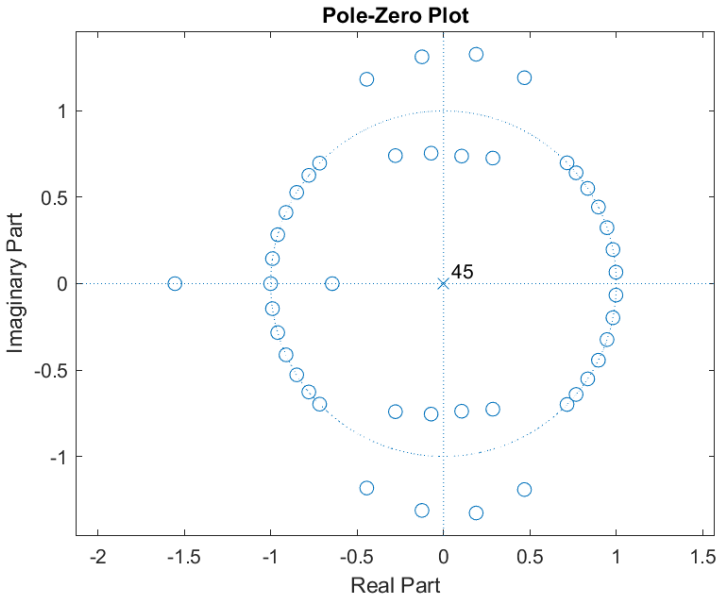


Figure 10: Pole zero plot of the windowed filter

equiripple

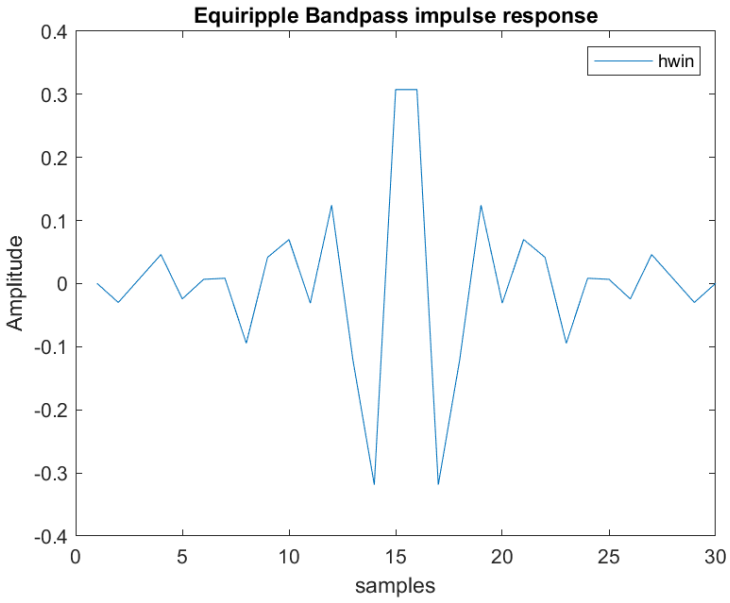


Figure 11: impulse response of the equiripple filter

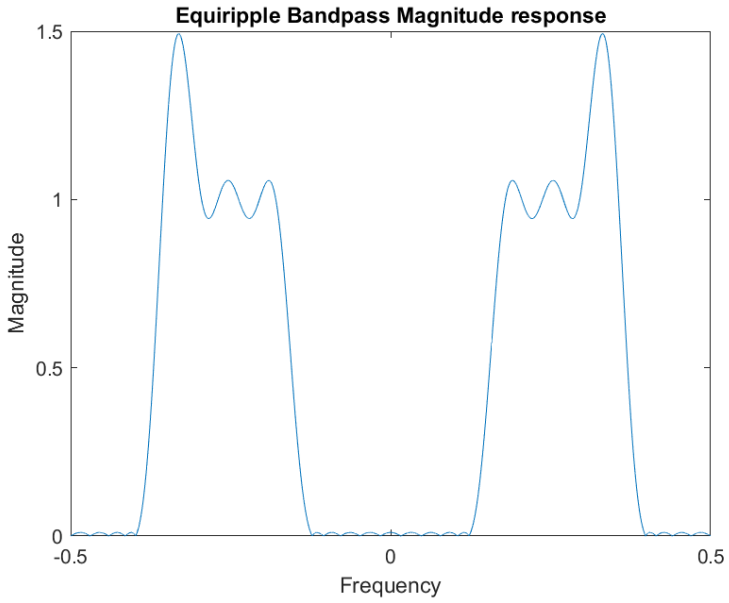


Figure 12: Linear magnitude response of the equiripple filter

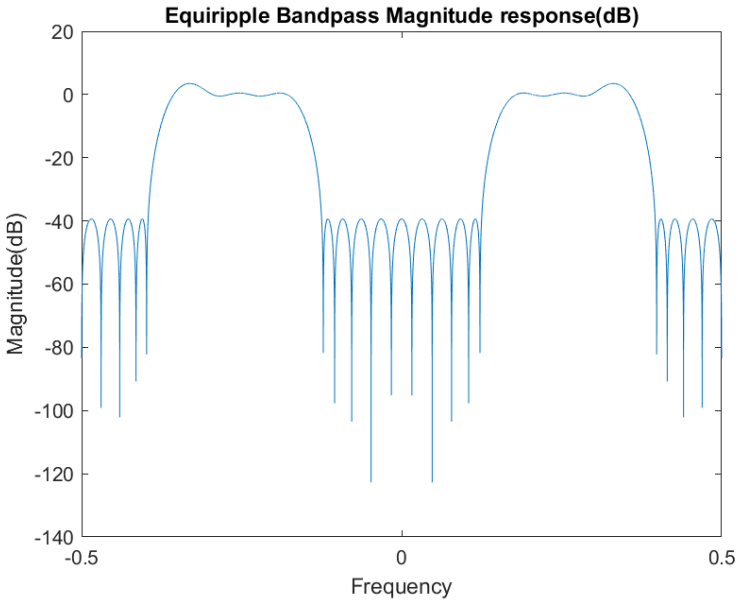


Figure 13: Logrithmic magnitude response of the equiripple filter

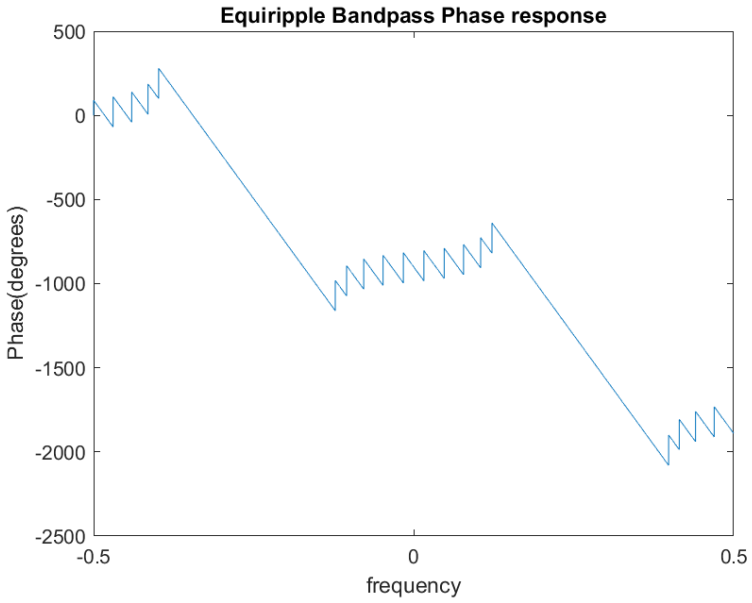


Figure 14: Phase response of the equiripple filter

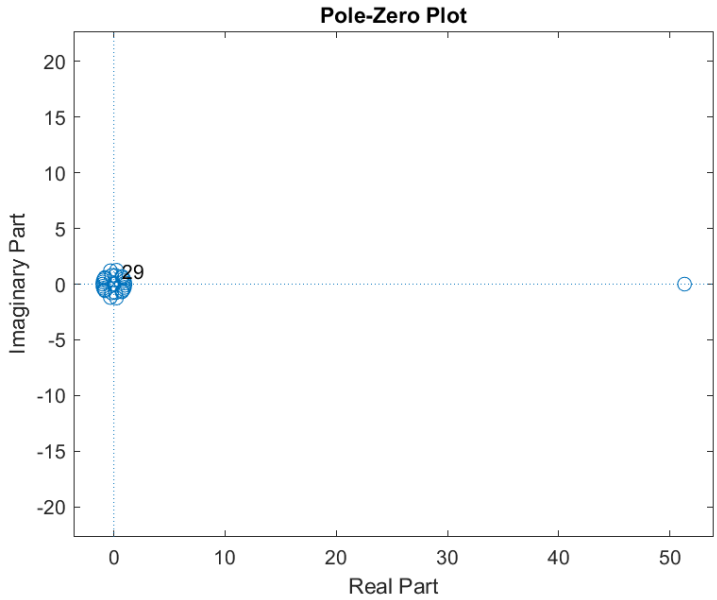


Figure 15: Pole zero plot of the equiripple filter

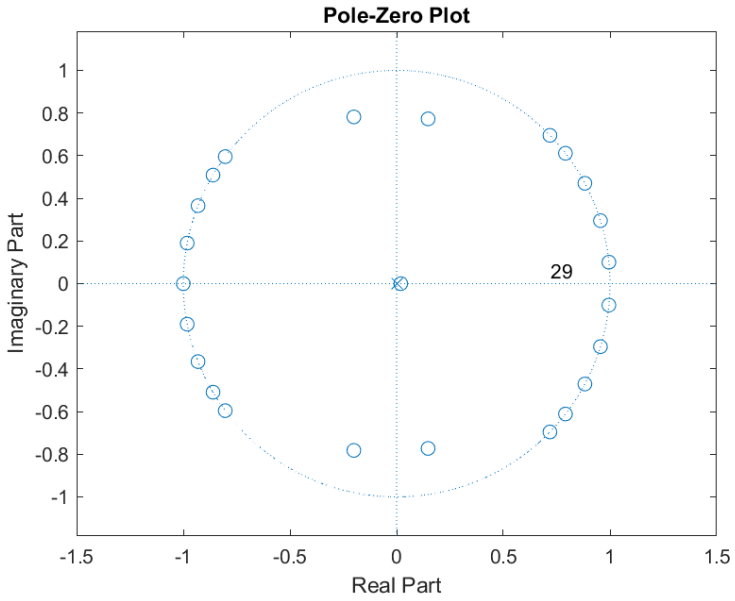


Figure 16: Pole zero plot of the equiripple filter about the unit circle

2. This problem explores designing a lowpass filter with a very narrow transition band that achieves the following specifications $A_p = 0.1$ db ripple in the passband ($0 \leq |f| \leq 0.24$) and $A_s = 60$dB suppression in the stopband ($0.26 \leq |f| \leq 0.5$).

filter design	Length	Order	group delay	type	window parameters
Window	275	274	137	I	Blackman
Kaiser	183	182	91	I	$\beta = 5.6533$
equiripple	137	136	68	I	equiripple

Window Design

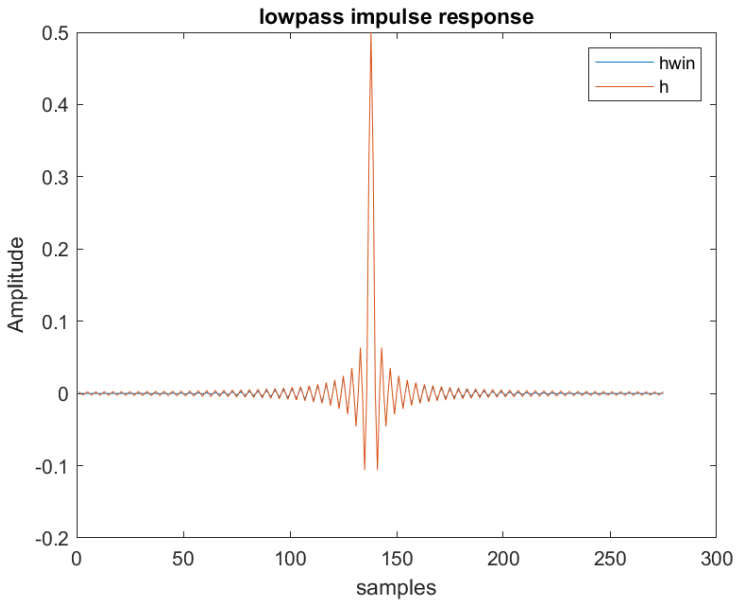


Figure 17: impulse response of the windowed and non windowed filter

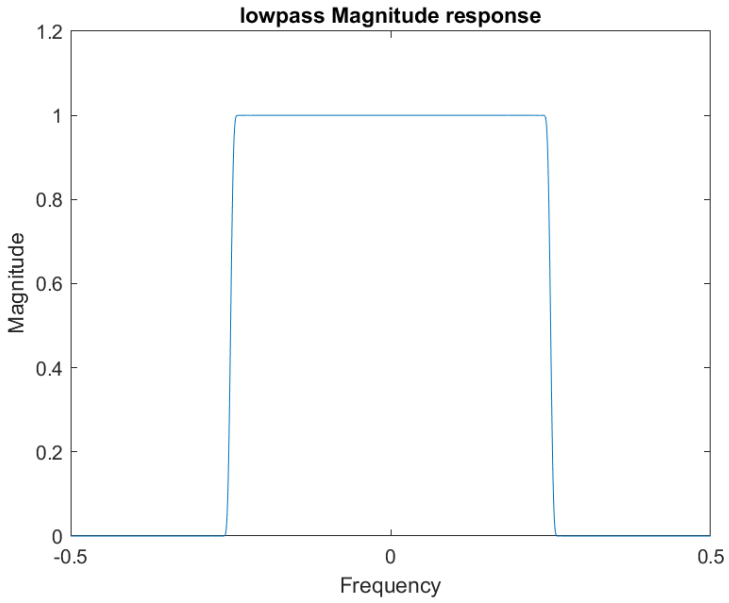


Figure 18: Linear magnitude response of the windowed filter

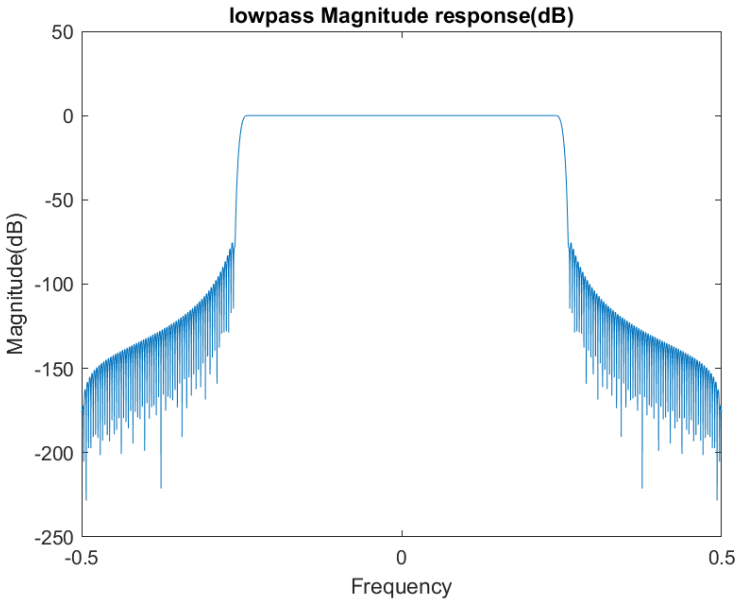


Figure 19: Logrithmic magnitude response of the windowed filter

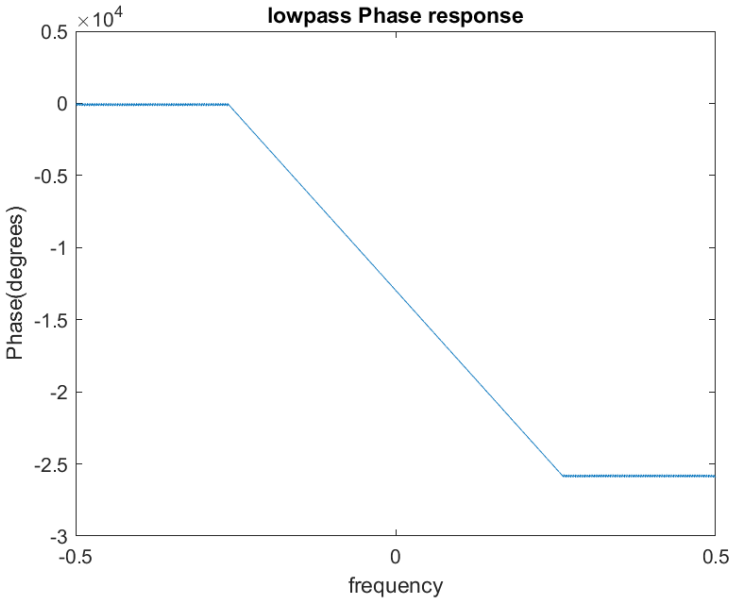


Figure 20: Phase response of the windowed filter

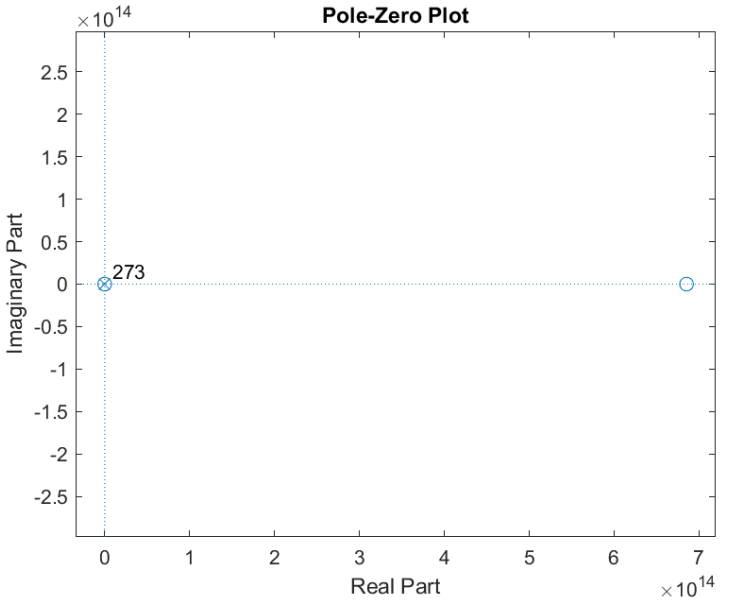


Figure 21: Pole zero plot of the windowed filter

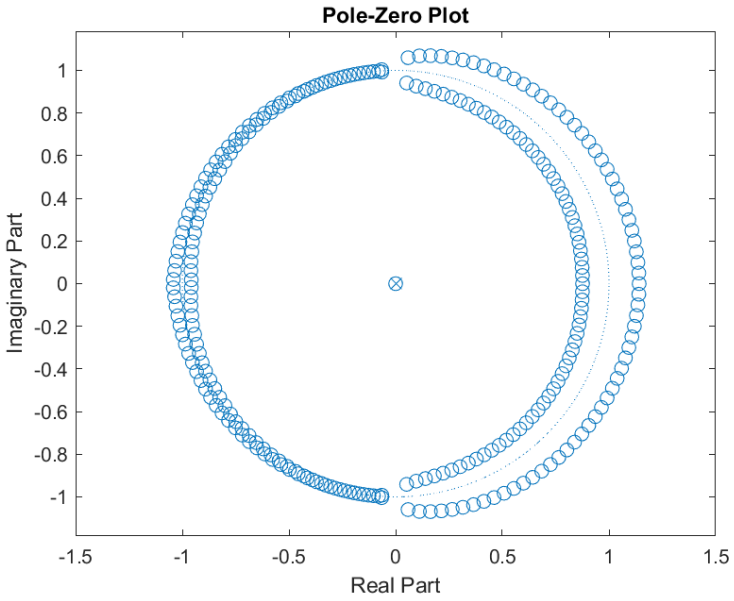


Figure 22: Pole zero plot of the windowed filter about the unit circle

Kaiser

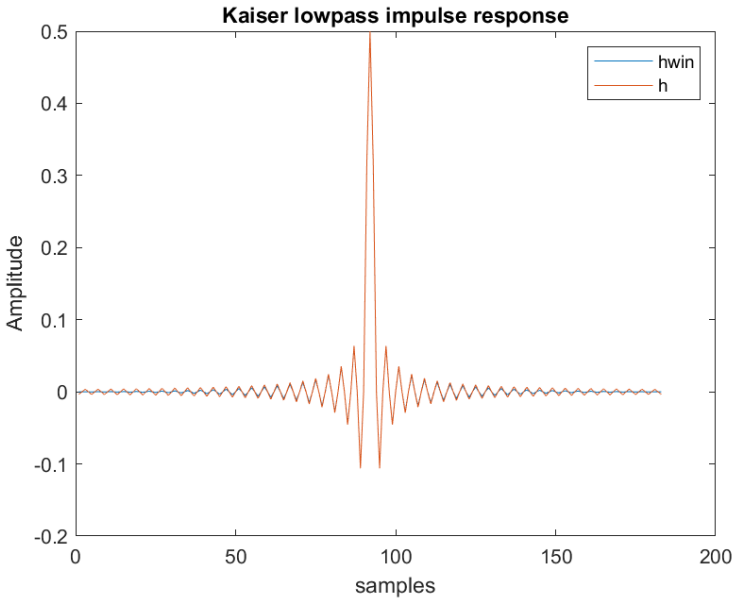


Figure 23: impulse response of the windowed and non windowed filter

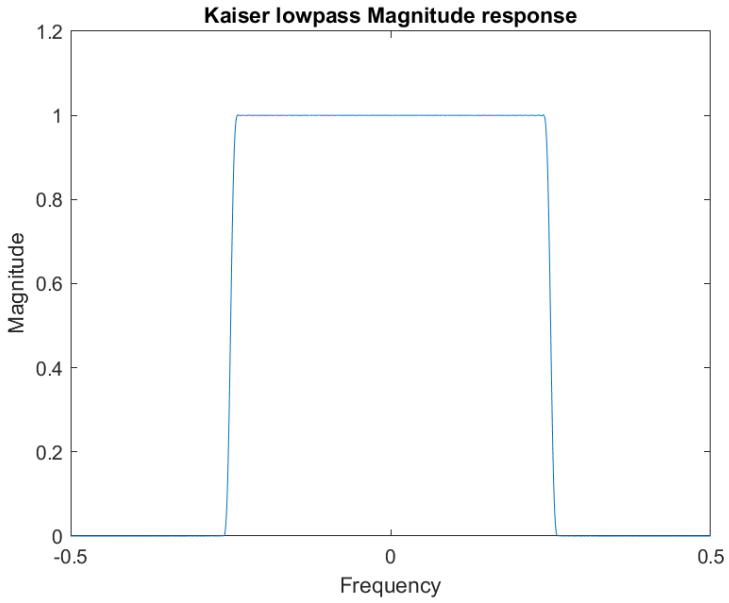


Figure 24: Linear magnitude response of the windowed filter

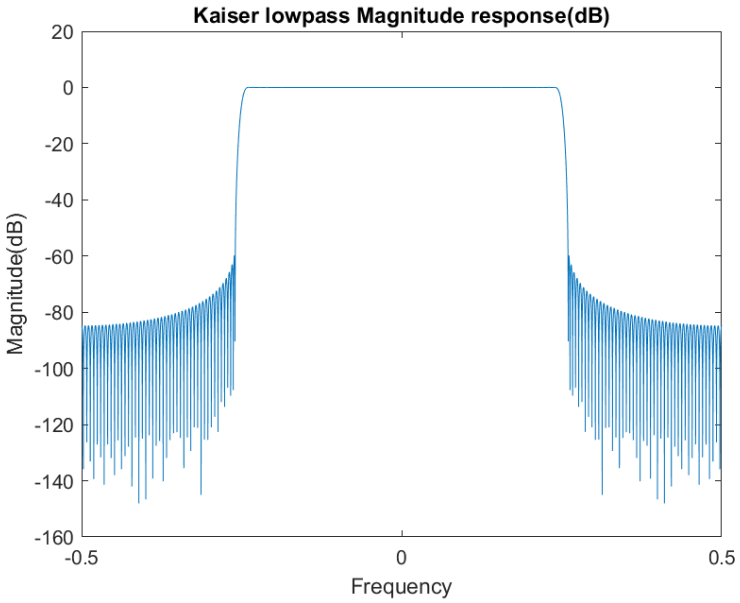


Figure 25: Logrithmic magnitude response of the windowed filter

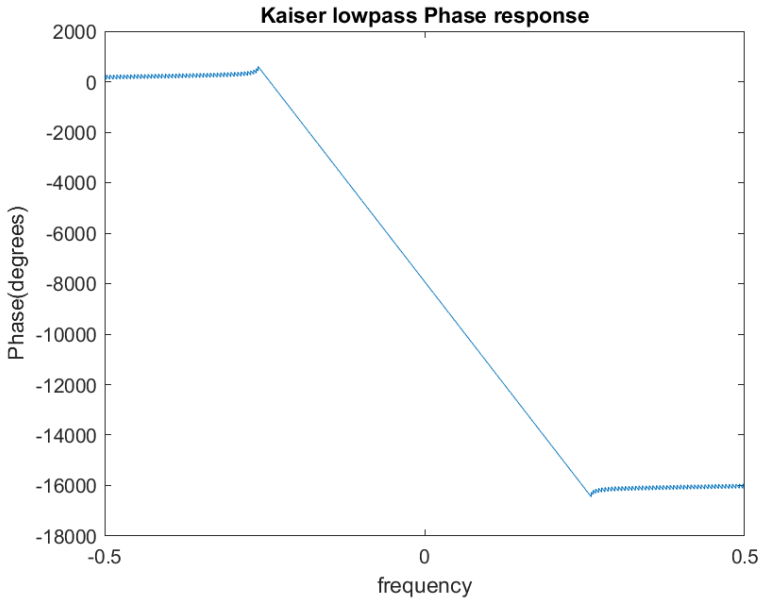


Figure 26: Phase response of the windowed filter

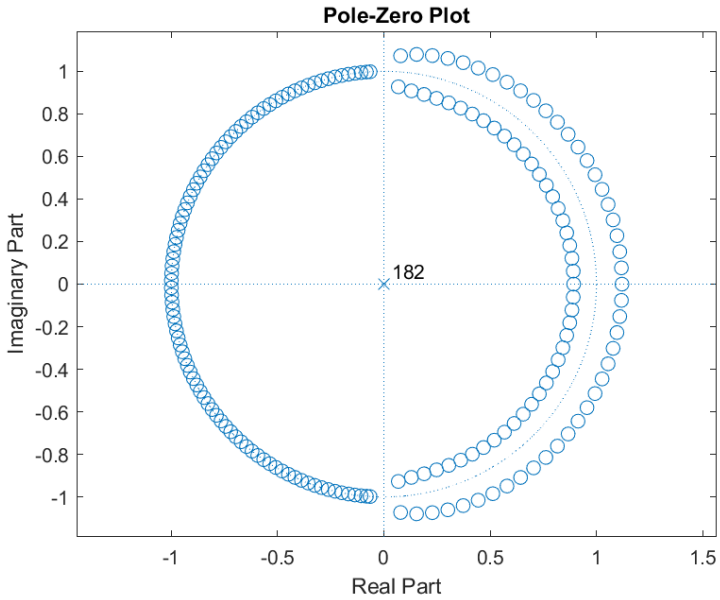


Figure 27: Pole zero plot of the windowed filter

equiripple

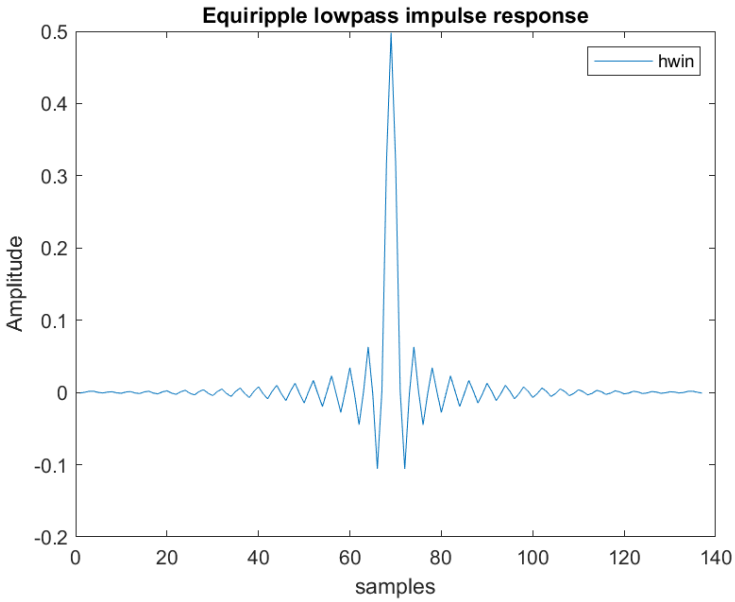


Figure 28: impulse response of the equiripple filter

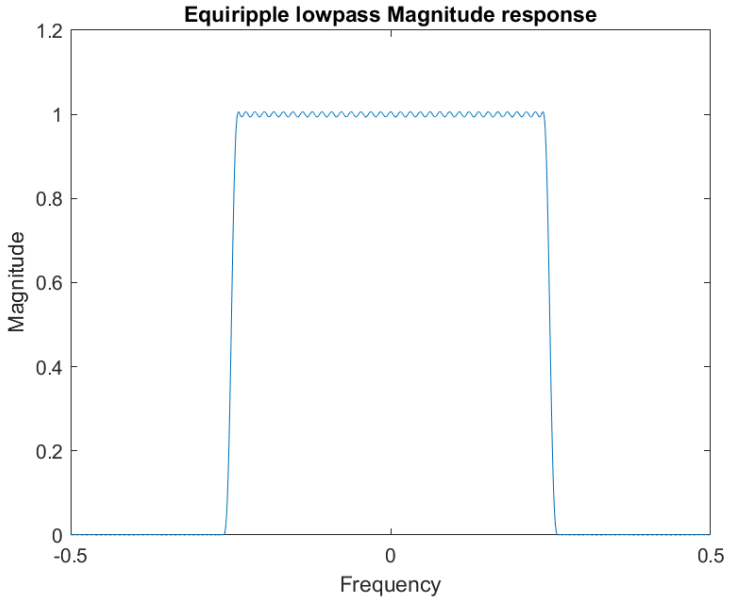


Figure 29: Linear magnitude response of the equiripple filter

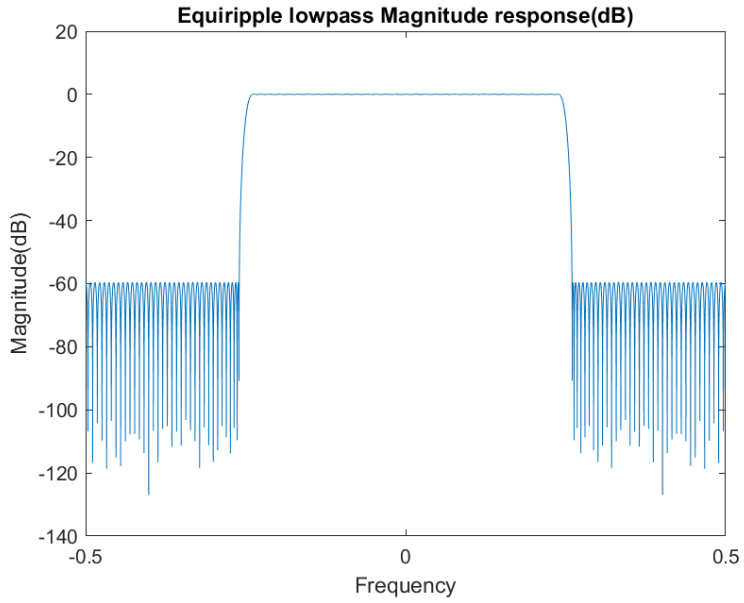


Figure 30: Logrithmic magnitude response of the equiripple filter

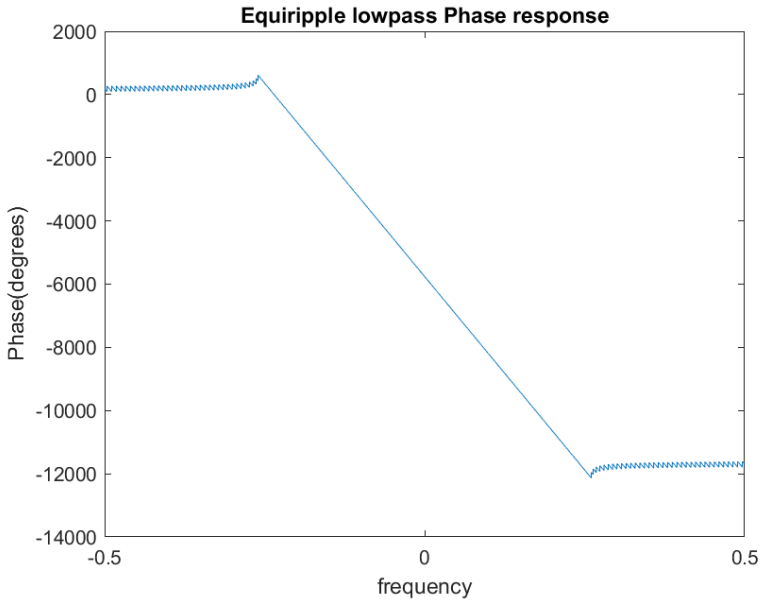


Figure 31: Phase response of the equiripple filter

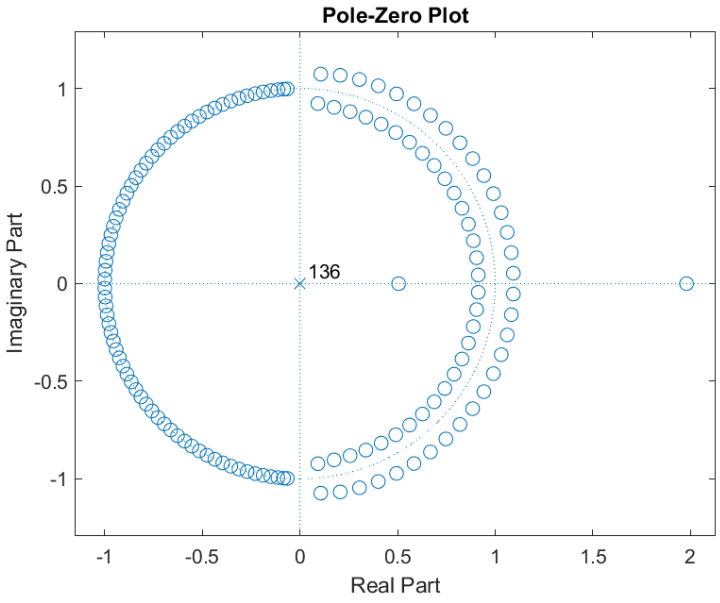


Figure 32: Pole zero plot of the equiripple filter

3. This problem explores designing a lowpass filter for sample rate conversion using the system depicted in Fig.6.47(c) in the textbook(page 379) in which the upsampling factor is U = 3 and the downsampling factor is D = 4. The maximum frequency in the signal x[n] is $f_{max,x} = 1/3$. design the lowpass filter to have $A_p = 0.1$ dB of ripple in the pass band and $A_s = 60$ dB of attenuation in the stopband. Use your knowledge of upsampling and downsampling to determine the passband and stopband edge frequencies, f_p and f_s. What are f_p and f_s.

$f_p = 1/9$ $f_s = 2/9$

filter design	Length	Order	group delay	type	window parameters
Window	50	49	24.5	II	Blackman
Kaiser	34	33	16.5	II	$\beta = 5.6533$
equiripple	25	24	12	I	equiripple

Window Design

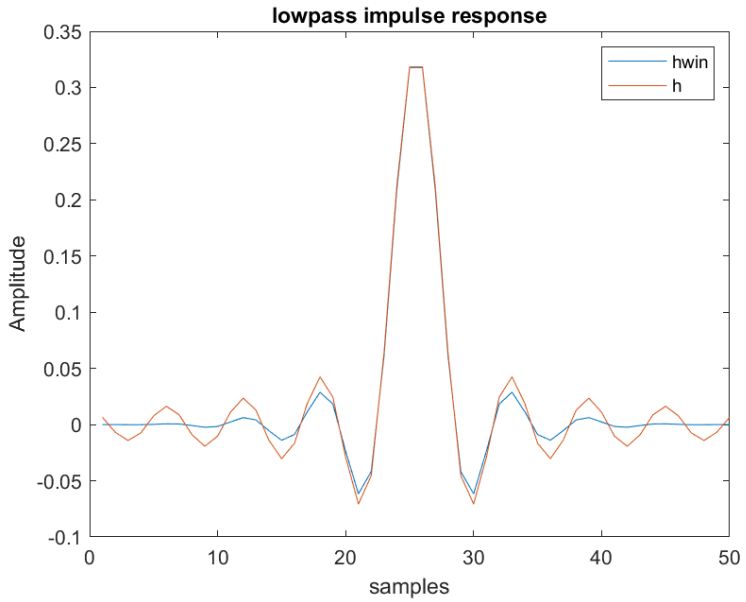


Figure 33: impulse response of the windowed and non windowed filter

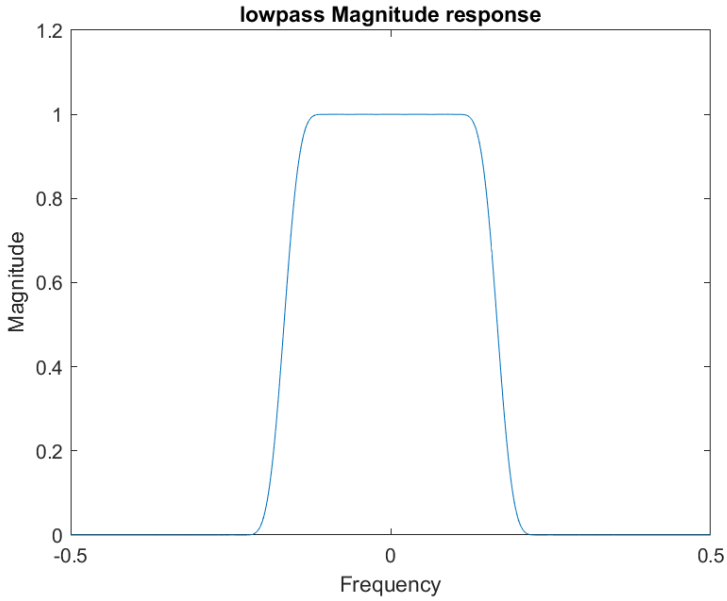


Figure 34: Linear magnitude response of the windowed filter

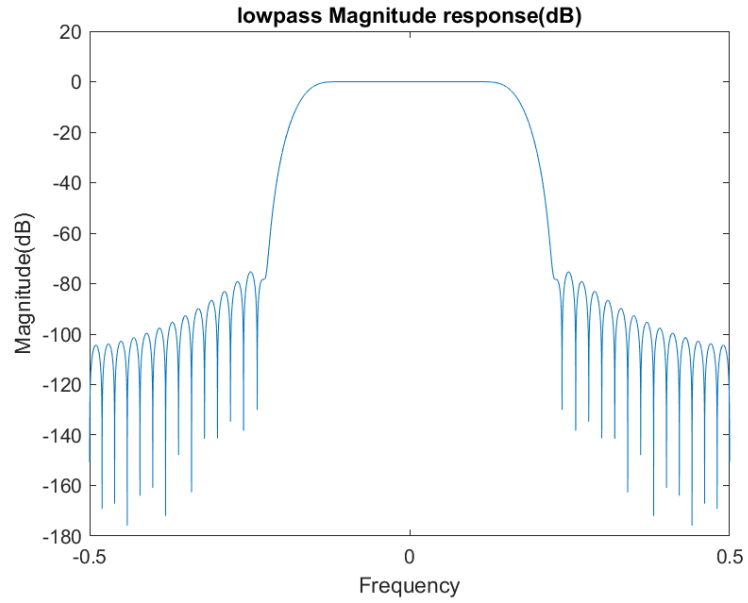


Figure 35: Logrithmic magnitude response of the windowed filter

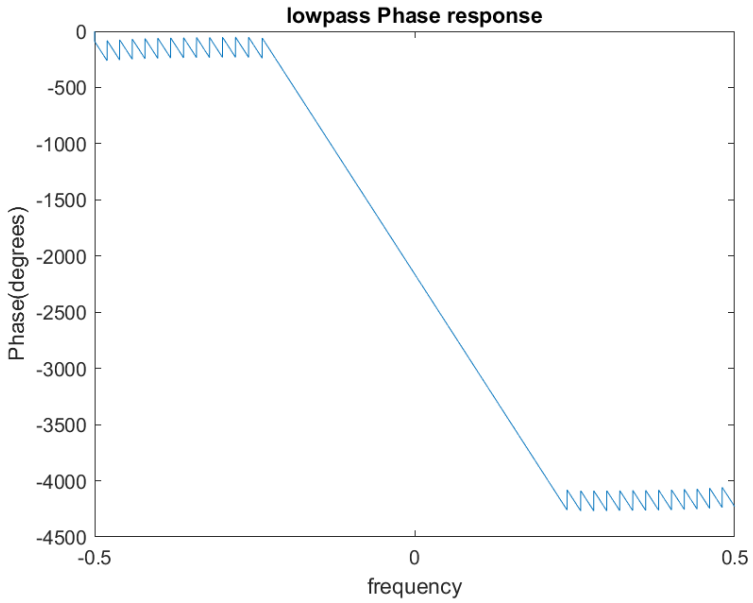


Figure 36: Phase response of the windowed filter

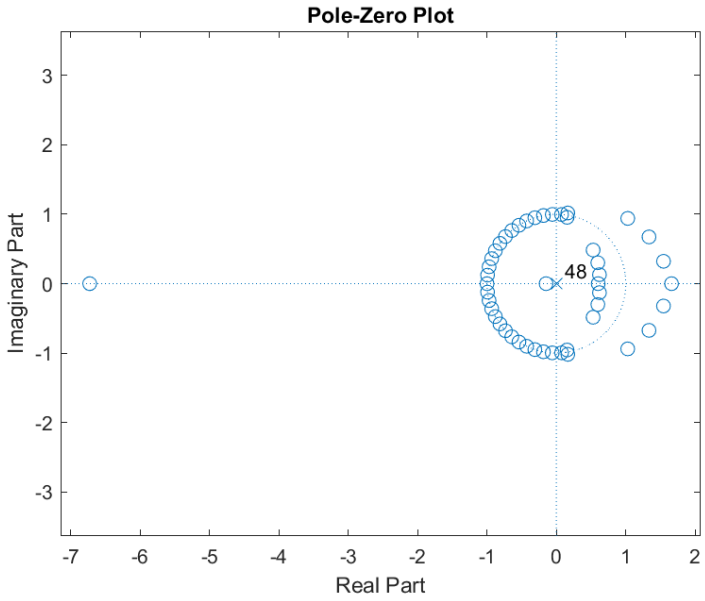


Figure 37: Pole zero plot of the windowed filter

Kaiser

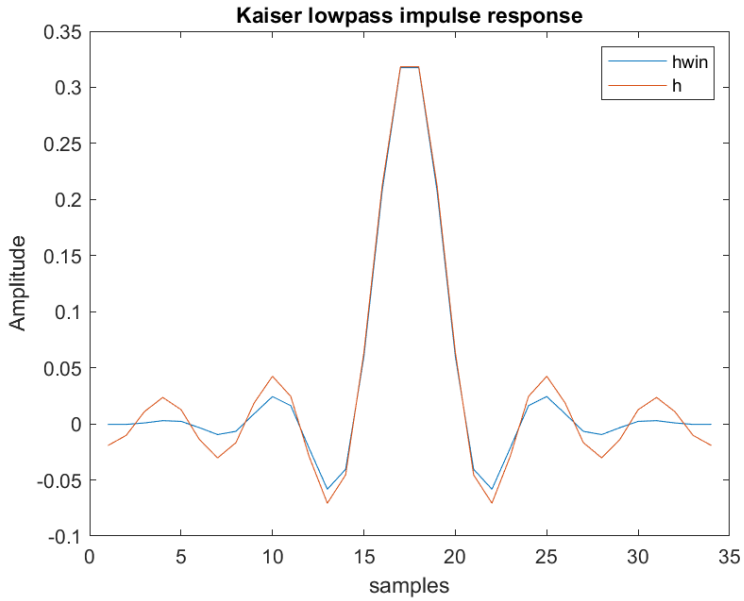


Figure 38: impulse response of the windowed and non windowed filter

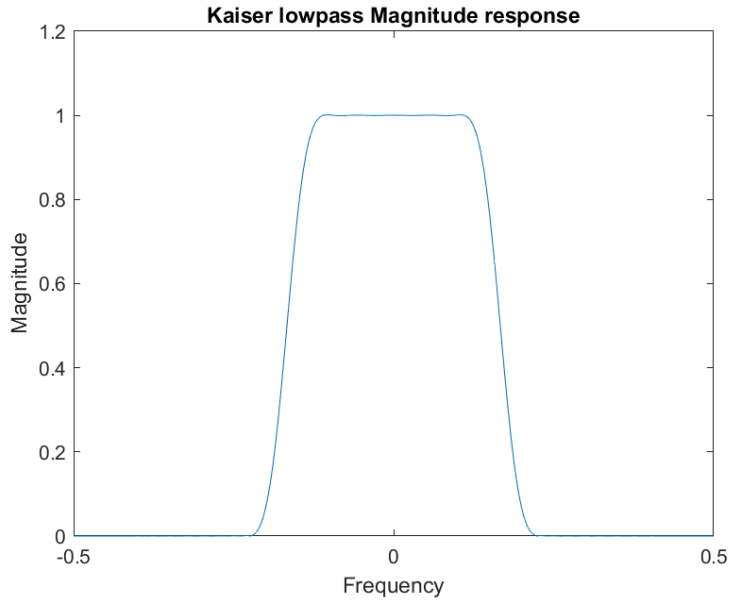


Figure 39: Linear magnitude response of the windowed filter

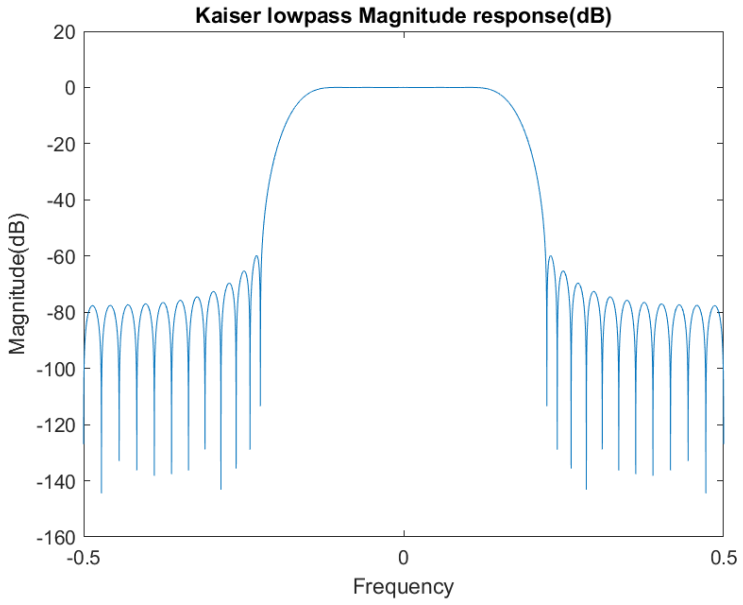


Figure 40: Logrithmic magnitude response of the windowed filter

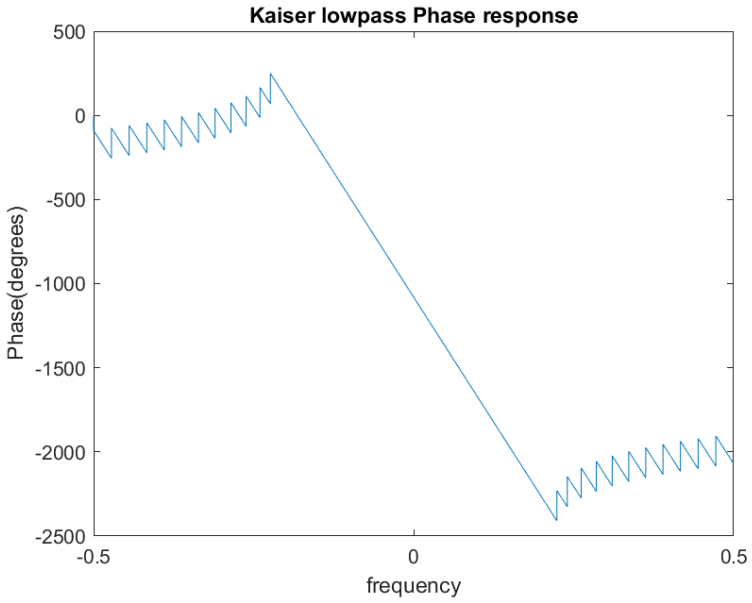


Figure 41: Phase response of the windowed filter

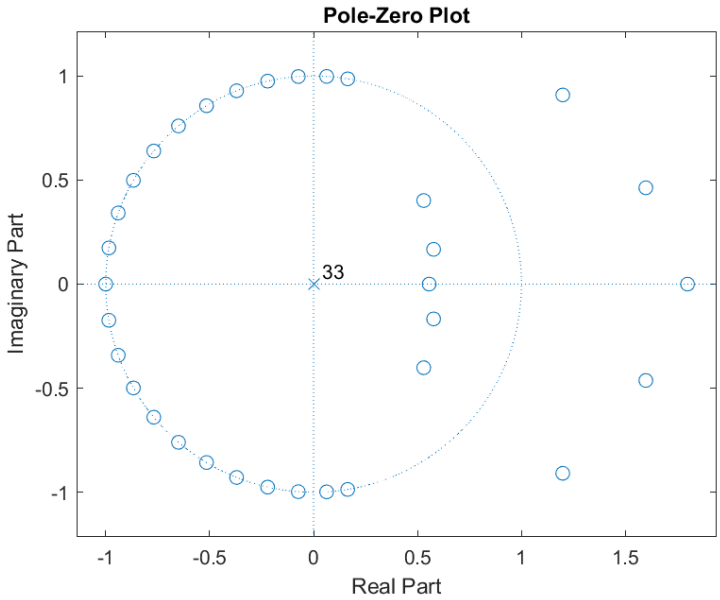


Figure 42: Pole zero plot of the windowed filter

equiripple

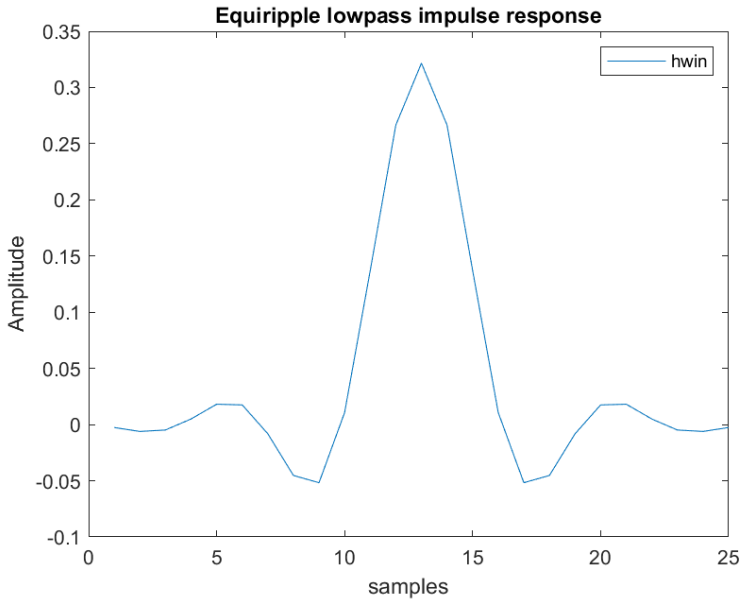


Figure 43: impulse response of the equiripple filter

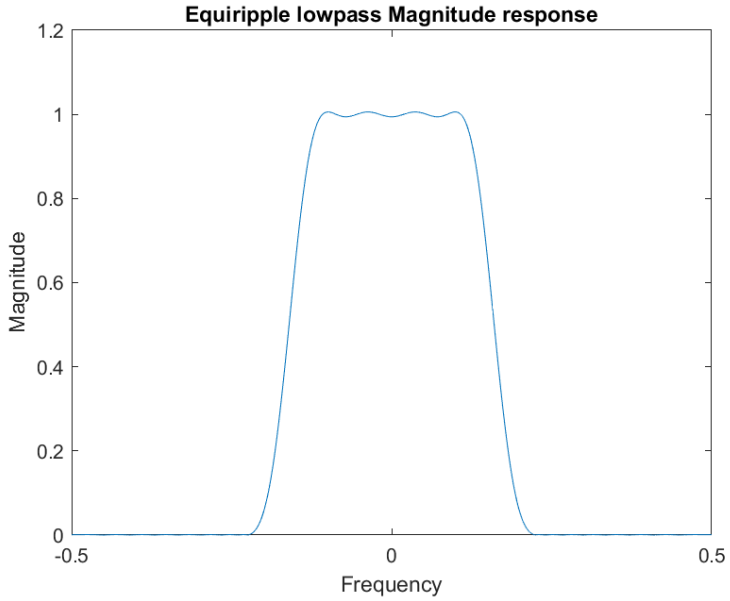


Figure 44: Linear magnitude response of the equiripple filter

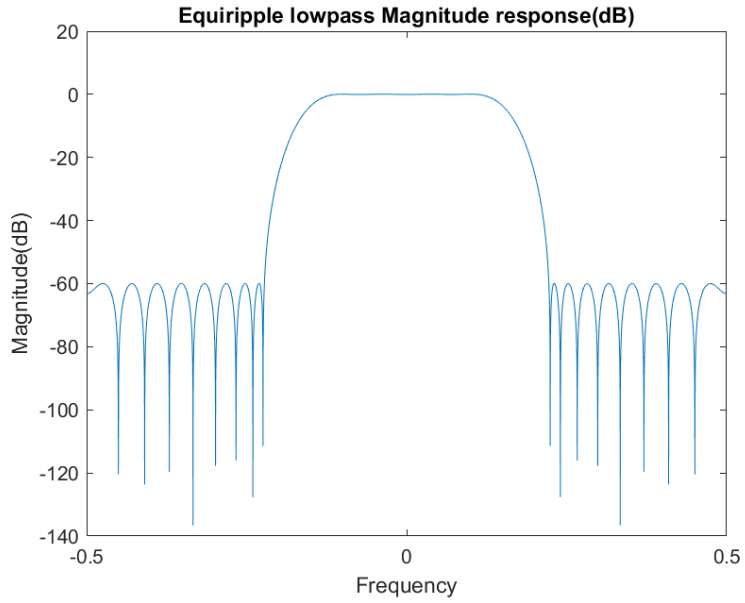


Figure 45: Logrithmic magnitude response of the equiripple filter

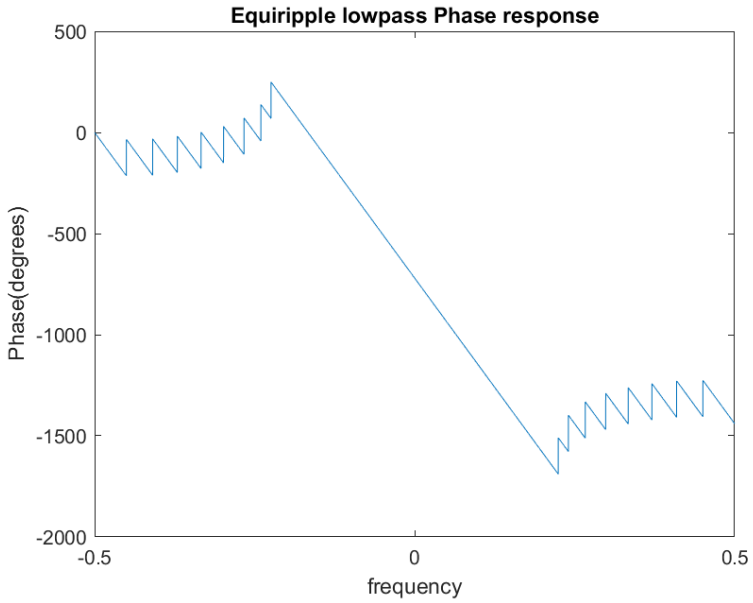


Figure 46: Phase response of the equiripple filter

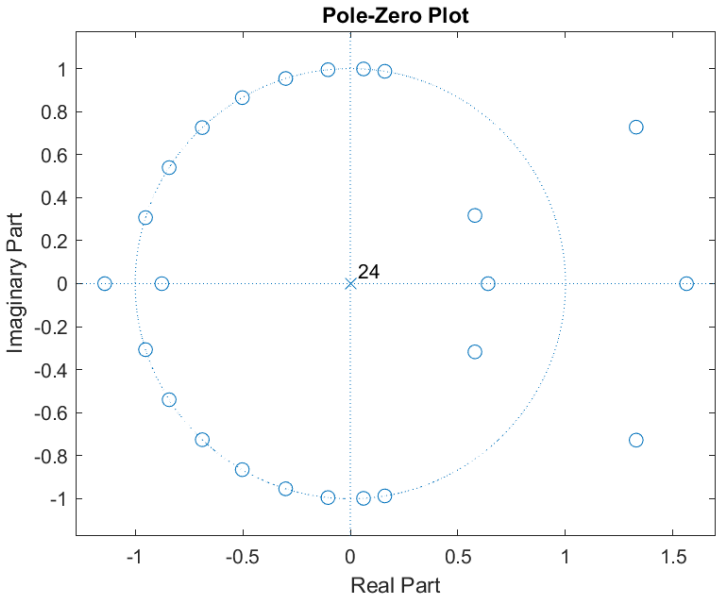


Figure 47: Pole zero plot of the equiripple filter

4. Suppose you would like to boost by a factor of 4 the low frequencies of an audio signal between the range of $0 \leq |\mathrm{F}| \leq 150$ Hz and leave the scale of frequencies above 300 Hz unchanged. This type of filter has two passbands and no stopband. is is called a shelving filter. The transtion band extends from 150 Hz to 300 Hz. You plan ot perfomr this processing using the system depected in Fig6.1 of the textbook (page 331) using a sample rate of $1/\mathrm{T} = \mathrm{F_s} = 20\mathrm{k}$ samples/second. Convert this information to a specification for a discrete-time filter and design a linear phase filter to meet this spec. Require your filter to limit deviations from the desired response to $\delta = 0.01$ in both pass bands.

filter design	Length	Order	group delay	type	window parameters
Window	441	440	220	I	Hamming
Kaiser	411	410	205	I	$\beta = 4.7717$
equiripple	353	352	176	I	equiripple

Window Design

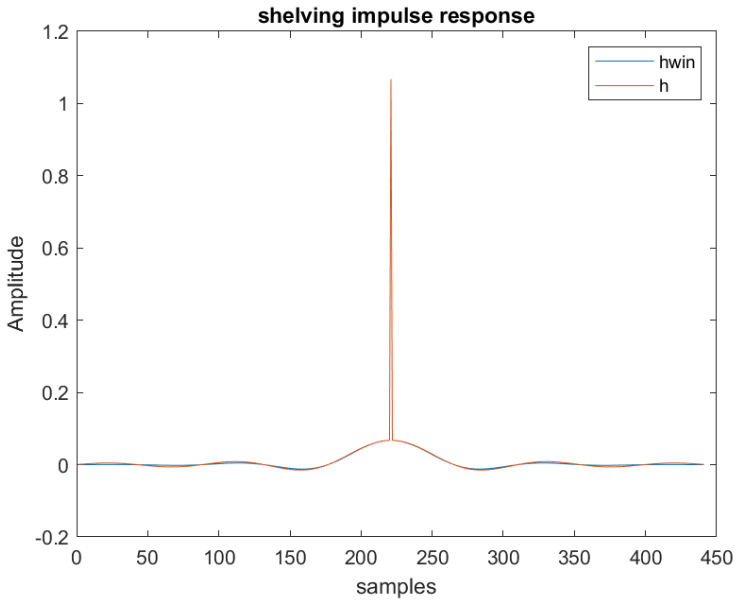


Figure 48: impulse response of the windowed and non windowed filter

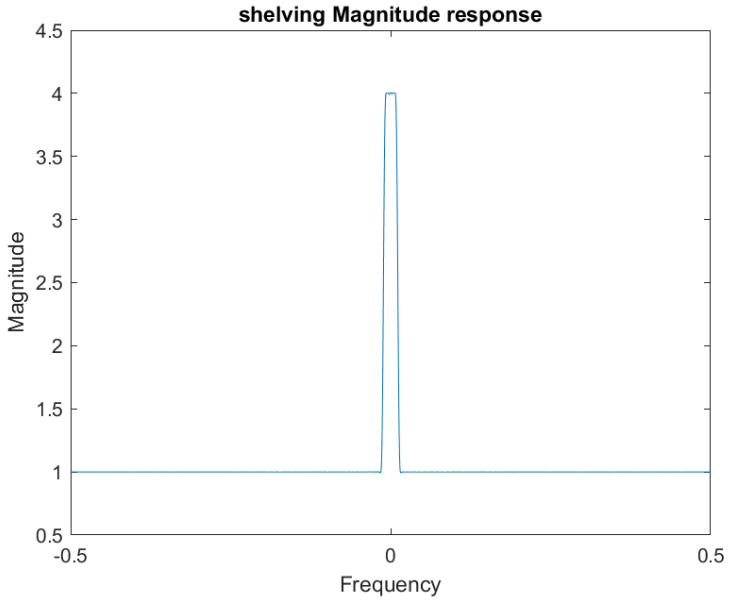


Figure 49: Linear magnitude response of the windowed filter

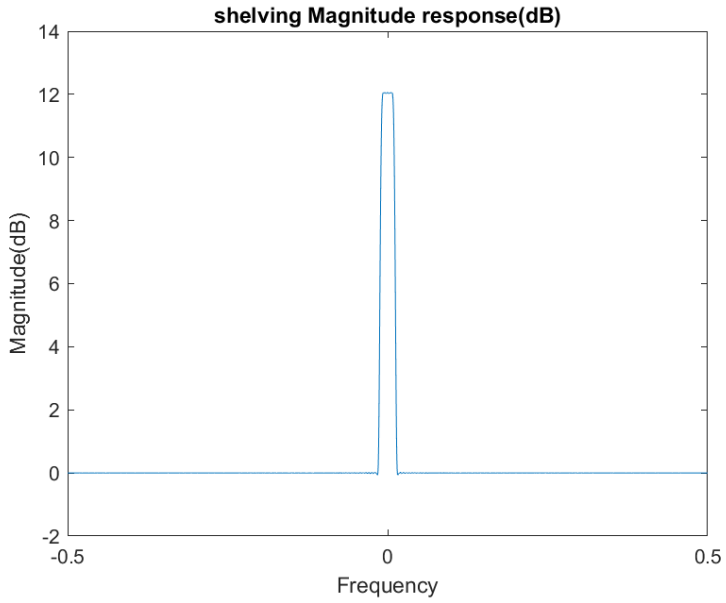


Figure 50: Logrithmic magnitude response of the windowed filter

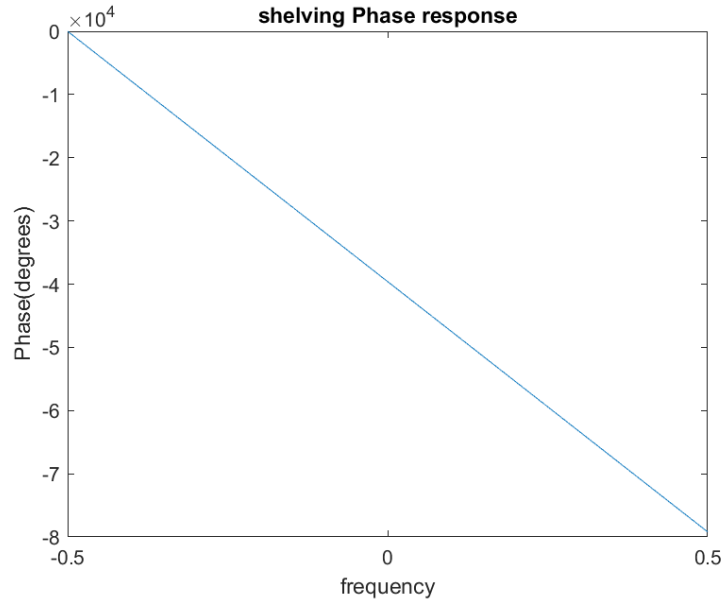


Figure 51: Phase response of the windowed filter

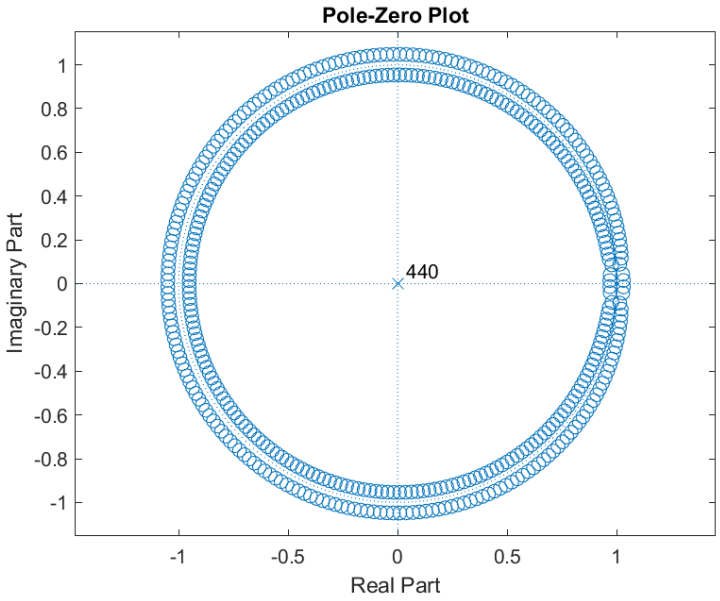


Figure 52: Pole zero plot of the windowed filter

Kaiser

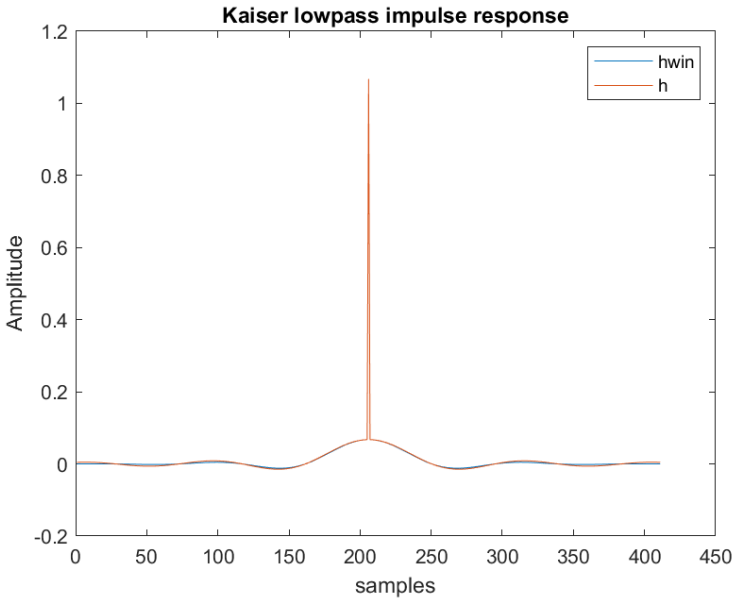


Figure 53: impulse response of the windowed and non windowed filter

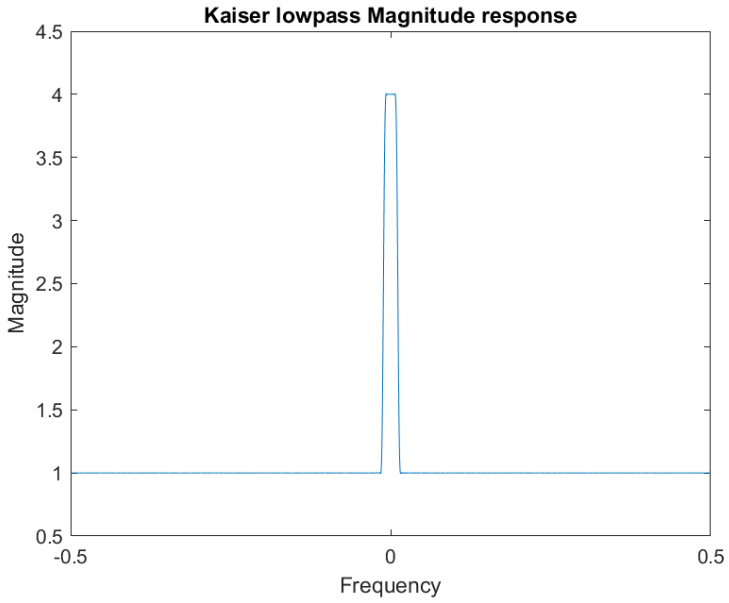


Figure 54: Linear magnitude response of the windowed filter

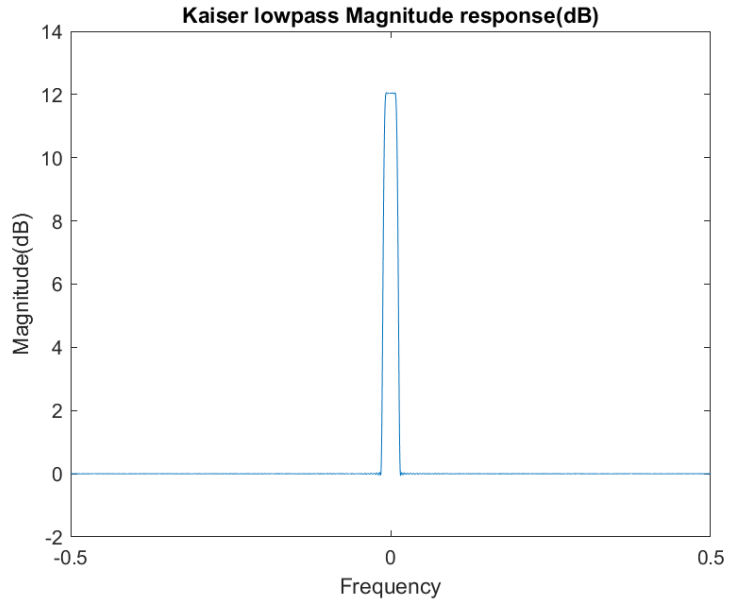


Figure 55: Logrithmic magnitude response of the windowed filter

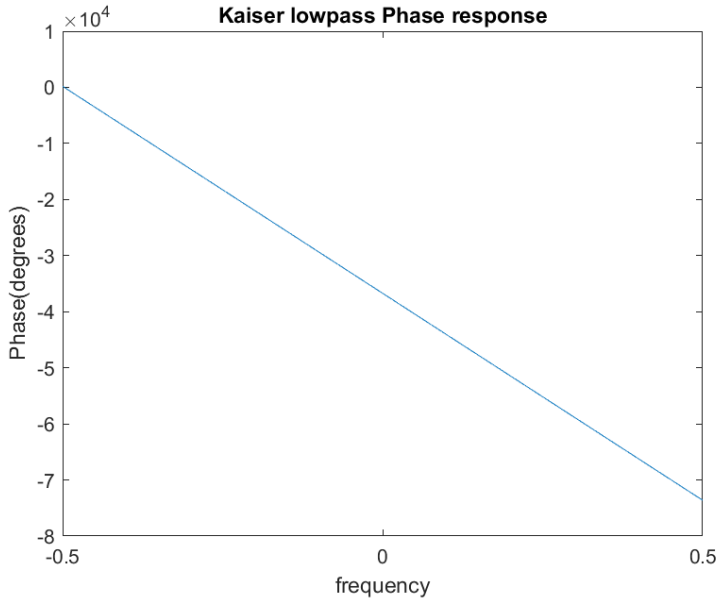


Figure 56: Phase response of the windowed filter

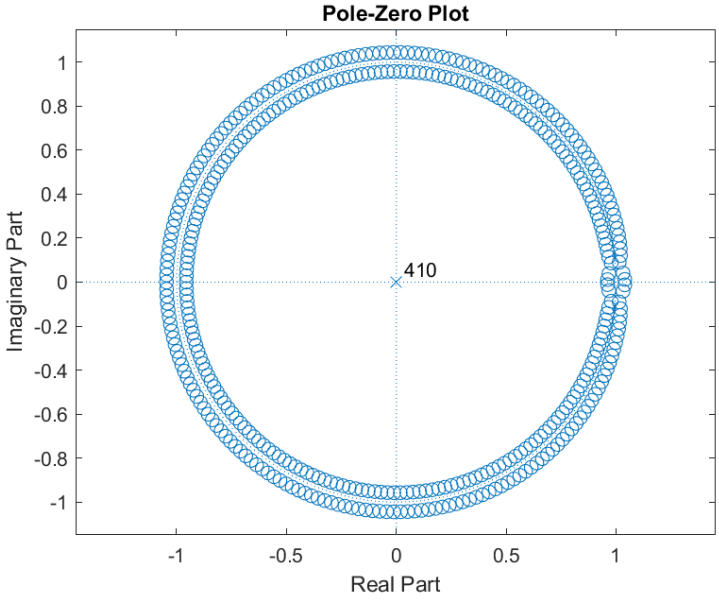


Figure 57: Pole zero plot of the windowed filter

equiripple

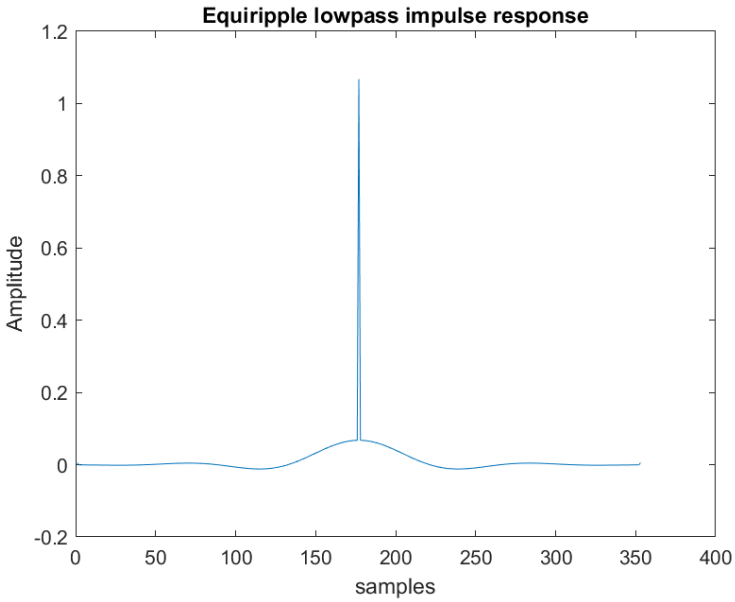


Figure 58: impulse response of the equiripple filter

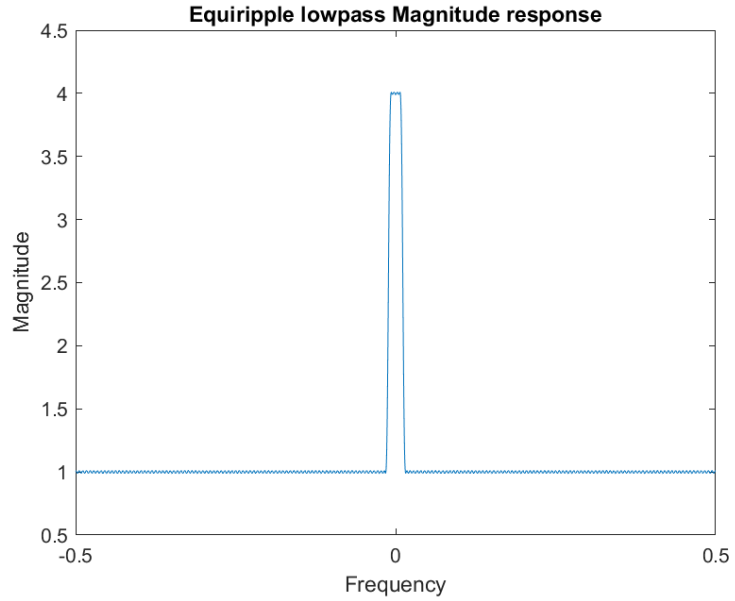


Figure 59: Linear magnitude response of the equiripple filter

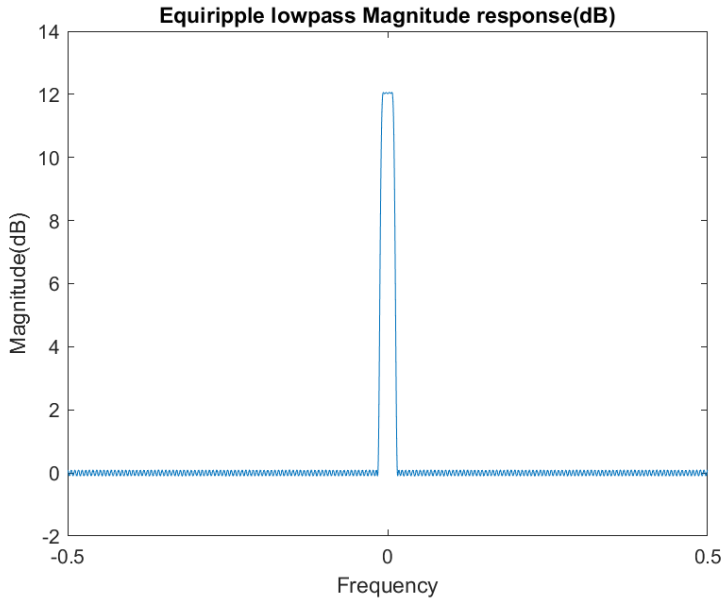


Figure 60: Logrithmic magnitude response of the equiripple filter

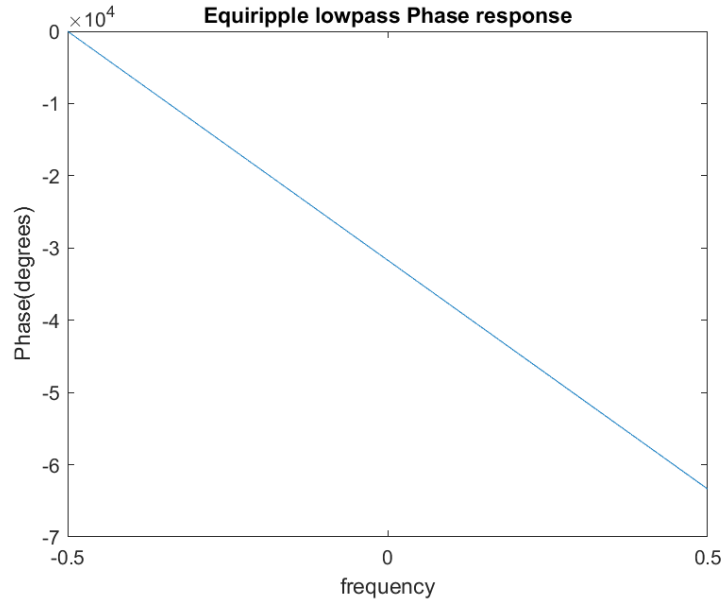


Figure 61: Phase response of the equiripple filter

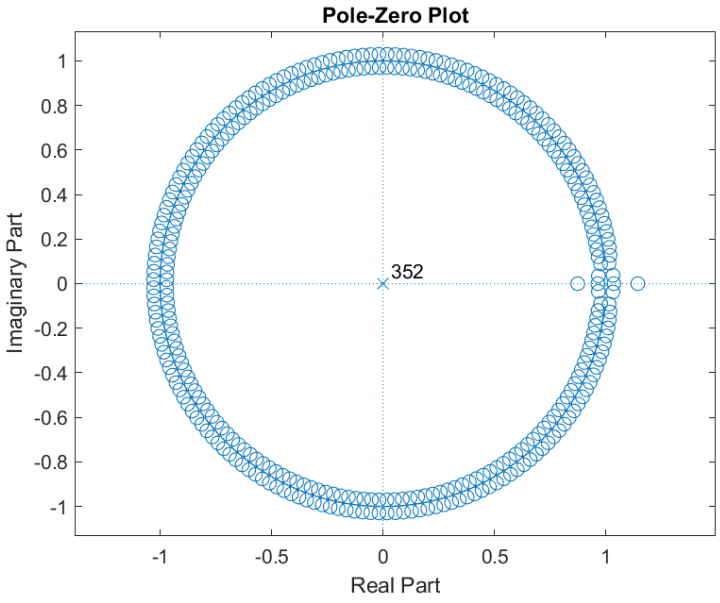


Figure 62: Pole zero plot of the equiripple filter

5. After working the previous 4 problems, summarize your observations about the best filter design method to use if the goal is to minimize the filter length.

Using the windowing technique to satisfy the parameters leads to over designing. The filters fufill the requirements but have higher order counts. Kaiser consistently has a 20 to 30 orders less than the window method. Equiripple has half to a third of the windowing order. Thus far equiripple design method is the best for minimizing order degree while fulfilling requirements.

6. Design a differentiator/delay filter pair to be used in FM demodulation. Let the received signal be x[n]. Design type III derivative and delay filters using a Hamming window with length L = 21. Design type IIII derivative and delay filters using a Hamming window with length L = 22. In each case, plot the impulse response and the frequency response over $-1/2 \leq f \leq 1/2$.

Type 3

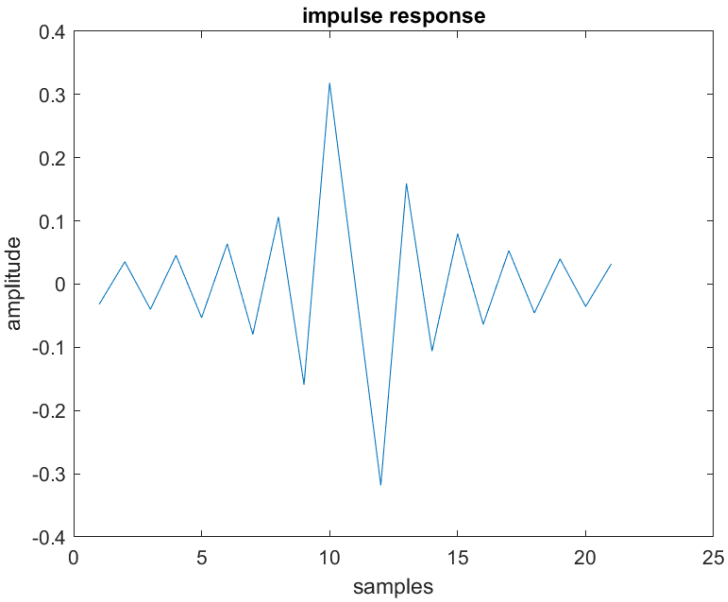


Figure 63: Impulse response of the type 3 difference filter.

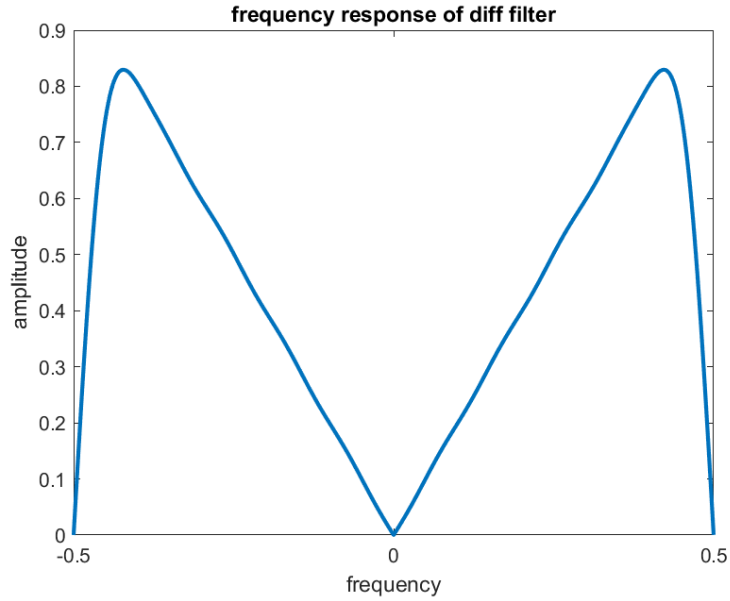


Figure 64: Frequency response of the type 3 difference filter.

Type 4

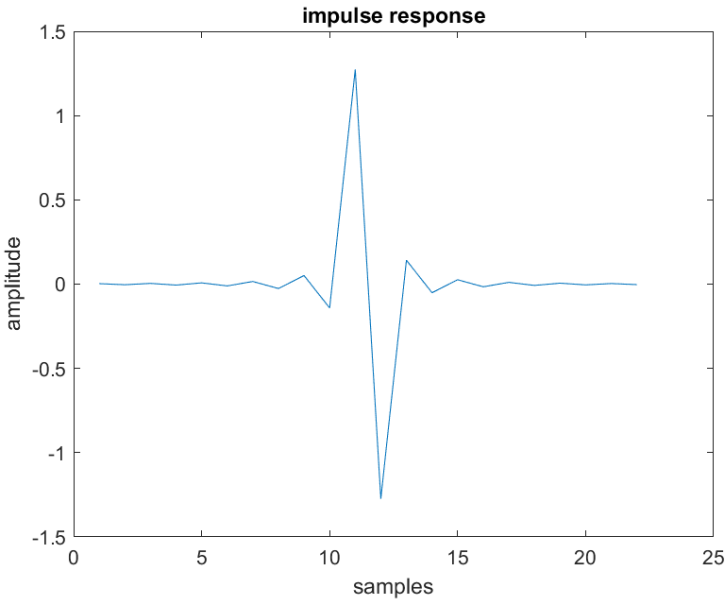


Figure 65: Impulse response of the type 4 difference filter.

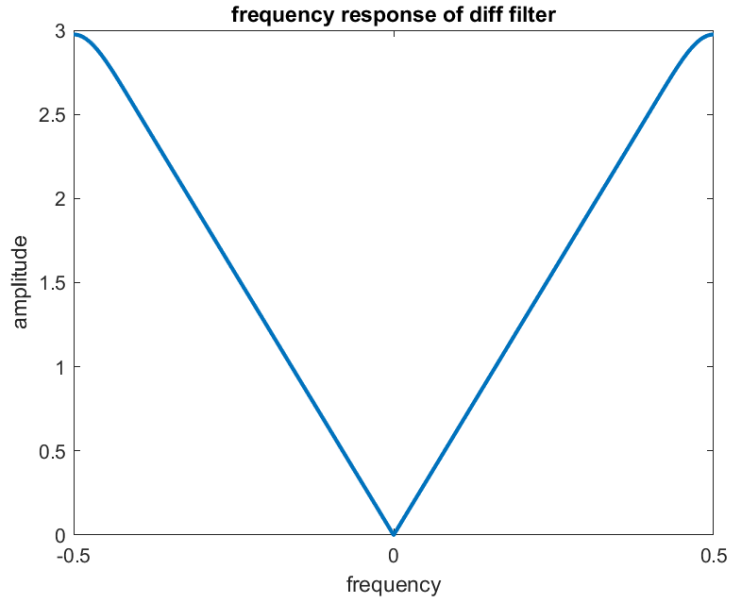


Figure 66: Frequency response of the type 4 difference filter.

Appendix

```
%% Problem 1
Ap = .87;
As = 40.424;
df = 0.05;
fc1 = .15;
fc2 = 0.35;

L = 62;
M = 61;
n = (0:M) - M/2;
N = 2^14;
win = hanning(L);
h = ((sin(2 * pi * fc2 .* n)./(pi .* n)) - (sin(2 * pi * fc1 .* n)./(pi .* n))).';
hwin = h .* win;
plot(hwin);
hold on;
plot(h);
title('Bandpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p1_impulse.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Bandpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p1_magresponse.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Bandpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p1_maglogresponse.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Bandpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p1_phaseresponse.png');
zplane(hwin.');
saveas(gcf,'p1_zplane.png');

beta = 0.5842 * (40-21)^0.4 + 0.07886 * (40 - 21);
M = ceil((40-8)/(2.285 * 2 * pi * .05));
L = M + 1;
n = (0:M) - M/2;
h = ((sin(2 * pi * fc2 .* n)./(pi .* n)) - (sin(2 * pi * fc1 .* n)./(pi .* n))).';
win = kaiser(L,beta);
hwin = h .* win;

plot(hwin);
hold on;
plot(h);
title('Kaiser Bandpass impulse response');
xlabel('samples');
```

```
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p1_impulse_k.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Kaiser Bandpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p1_magresponse_k.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Kaiser Bandpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p1_maglogresponse_k.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Kaiser Bandpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p1_phaseresponse_k.png');
zplane(hwin.');
saveas(gcf,'p1_zplane_k.png');

%equiripple
h = load('p1_equiripple.fcf');
plot(h);
title('Equiripple Bandpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p1_impulse_eq.png');
Hwin = fft(h,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Equiripple Bandpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p1_magresponse_eq.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Equiripple Bandpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p1_maglogresponse_eq.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Equiripple Bandpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p1_phaseresponse_eq.png');
zplane(h.');
saveas(gcf,'p1_zplane_eq.png');
xlim([-1.5 1.5]);
saveas(gcf,'p1_zplane_eq_zoomed.png');

%% problem 2
Ap = .1;
As = 60;
```

```
gamma = 10^(Ap/20);
deltap = (gamma-1)/(gamma + 1);
deltas = 10^(-As/20);
delta = min([deltap deltas]);
A = 20 * log10(delta);
fp = .24;
fs = .26;
df = fs-fp;
fc = .5*(fs + fp);

%blackman
L = ceil((11)/(2 * df));
M = L - 1;
N = 2^15;
win = blackman(L).';
n = (0:M) - M/2;
h = (2 * fc)*sinc(2 * fc .* n);
hwin = h .* win;

plot(hwin);
hold on;
plot(h);
title('lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p2_impulse.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('lowpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p2_magresponse.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p2_maglogresponse.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p2_phaseresponse.png');
zplane(hwin);
saveas(gcf,'p2_zplane.png');
xlim([-1.5 1.5]);
saveas(gcf,'p2_zplane_zoomed.png');

%Kaiser
gamma = 10^(Ap/20)
dp = (gamma - 1)/(gamma + 1);
ds = 10^(-As/20);
delta = min([dp ds]);
A = -20*log10(delta);
```

```
beta = 0.1102 * (A - 8.7);
M = ceil((A-8)/(2.285 * 2 * pi * df))
L = M + 1;
n = (0:M) - M/2;
h = (2 * fc)*sinc(2 * fc .* n);
win = kaiser(L,beta).';
hwin = h .* win;

plot(hwin);
hold on;
plot(h);
title('Kaiser lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p2_impulse_k.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Kaiser lowpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p2_magresponse_k.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Kaiser lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p2_maglogresponse_k.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Kaiser lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p2_phaseresponse_k.png');
zplane(hwin);
saveas(gcf,'p2_zplane_k.png');

%equiripple
h = load('p2_equiripple.fcf');
plot(h);
title('Equiripple lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p2_impulse_eq.png');
Hwin = fft(h,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Equiripple lowpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p2_magresponse_eq.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Equiripple lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p2_maglogresponse_eq.png');
```

```
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Equiripple lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p2_phaseresponse_eq.png');
zplane(h.');
saveas(gcf,'p2_zplane_eq.png');
xlim([-1.5 1.5]);
saveas(gcf,'p2_zplane_eq_zoomed.png');

%% Problem 3
Ap = .1;
As = 60;
fp = 1/9;
fs = 2/9;
df = fs-fp;
fc = .5*(fs + fp);

%blackman
L = ceil((11 * pi)/(2 * pi * df));
M = L - 1;
N = 2^15;
win = blackman(L).';
n = (0:M) - M/2;
h = (2 * fc)*sinc(fc * 2 .* n);
hwin = h .* win;

plot(hwin);
hold on;
plot(h);
title('lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p3_impulse.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('lowpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p3_magresponse.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p3_maglogresponse.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p3_phaseresponse.png');
zplane(hwin);
saveas(gcf,'p3_zplane.png');
xlim([-1.5 1.5]);
saveas(gcf,'p3_zplane_zoomed.png');
```

```
gamma = 10^(Ap/20)
dp = (gamma - 1)/(gamma + 1);
ds = 10^(-As/20);
delta = min([dp ds]);
A = -20*log10(delta);

beta = 0.1102 * (A - 8.7);
M = ceil((A-8)/(2.285 * 2 * pi * df));
L = M + 1;
n = (0:M) - M/2;
h = (2 * fc)*sinc(fc * 2 .* n);
win = kaiser(L,beta).';
hwin = h .* win;

plot(hwin);
hold on;
plot(h);
title('Kaiser lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p3_impulse_k.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Kaiser lowpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p3_magresponse_k.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Kaiser lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p3_maglogresponse_k.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Kaiser lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p3_phaseresponse_k.png');
zplane(hwin);
saveas(gcf,'p3_zplane_k.png');

%equiripple
h = load('p3_equiripple.fcf');
plot(h);
title('Equiripple lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p3_impulse_eq.png');
Hwin = fft(h,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Equiripple lowpass Magnitude response');
xlabel('Frequency');
```

```
ylabel('Magnitude')
saveas(gcf,'p3_magresponse_eq.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Equiripple lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p3_maglogresponse_eq.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Equiripple lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p3_phaseresponse_eq.png');
zplane(h.');
saveas(gcf,'p3_zplane_eq.png');
xlim([-1.5 1.5]);
saveas(gcf,'p3_zplane_eq_zoomed.png');

%% Problem 4
delta = 0.01;
Ap = 20*log10((1 + delta/3)/(1-delta/3));
As = 20 * log10((1 + delta/3)/(delta/3));

Fs = 20e3;
fp = 150/Fs;
fs = 300/Fs;
df = fs-fp;
fc = .5*(fs + fp);

%blackman
L = ceil((6.6 * pi)/(2 * pi * df))+1;
M = L - 1;
N = 2^15;
win = hamming(L).';
n = (0:M) - M/2;
h1 = 3*(2 * fc)*sinc(fc * 2 .* n);
h2 = sinc(n);
h = h1 + h2;
hwin = h .* win;

plot(hwin);
hold on;
plot(h);
title('shelving impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p4_impulse.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('shelving Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p4_magresponse.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('shelving Magnitude response(dB)')
```

```
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p4_maglogresponse.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('shelving Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p4_phaseresponse.png');
zplane(hwin);
saveas(gcf,'p4_zplane.png');

%Kaiser
gamma = 10^(Ap/20)
dp = (gamma - 1)/(gamma + 1);
ds = 10^(-As/20);
delta = min([dp ds]);
A = 52;

%beta = 0.5842*(A-21)^0.4 + 0.07886*(A-21);
beta = 0.1102* (A - 8.7);
M = ceil((A-8)/(2.285 * 2 * pi * df)) + 1;
L = M + 1;
n = (0:M) - M/2;

h1 = 3*(2 * fc)*sinc(fc * 2 .* n);
%Change delay filter to a sinc function
h2 = sinc(n);
h = h1 + h2;
win = kaiser(L,beta).';
hwin = h .* win;

plot(hwin);
hold on;
plot(h);
title('Kaiser lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
legend('hwin', 'h');
hold off
saveas(gcf, 'p4_impulse_k.png');
Hwin = fft(hwin,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Kaiser lowpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p4_magresponse_k.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Kaiser lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p4_maglogresponse_k.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Kaiser lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p4_phaseresponse_k.png');
```

```
zplane(hwin);
saveas(gcf,'p4_zplane_k.png');

%equiripple
h = 3 * load('p4_equiripple.fcf').';
M = length(h) - 1;
L = length(h);
h(M/2+1) = h(M/2 + 1) + 1;

plot(h);
title('Equiripple lowpass impulse response');
xlabel('samples');
ylabel('Amplitude');
hold off
saveas(gcf, 'p4_impulse_eq.png');
Hwin = fft(h,N);
plot((0:N-1)/N - .5,abs(fftshift(Hwin)));
title('Equiripple lowpass Magnitude response');
xlabel('Frequency');
ylabel('Magnitude')
saveas(gcf,'p4_magresponse_eq.png');
plot((0:N-1)/N - .5,20*log10(abs(fftshift(Hwin))));
title('Equiripple lowpass Magnitude response(dB)')
xlabel('Frequency');
ylabel('Magnitude(dB)');
saveas(gcf,'p4_maglogresponse_eq.png');
plot((0:N-1)/N - .5,rad2deg(unwrap(angle(fftshift(Hwin)))));
title('Equiripple lowpass Phase response');
xlabel('frequency');
ylabel('Phase(degrees)')
saveas(gcf,'p4_phaseresponse_eq.png');
zplane(h);
saveas(gcf,'p4_zplane_eq.png');
xlim([-1.5 1.5]);
saveas(gcf,'p4_zplane_eq_zoomed.png');

%% Problem 6
%type 3 filter
L = 21;
M = 20;
n = (0:M) - M/2;
h = cos(pi .* n)./(pi .* n) - sin(pi .* n)./(pi .* n.^2);
h(M/2 + 1) = 0;

w = hamming(L).';
d = h.*w;

NFFT = 2^16;
f = [-NFFT/2:NFFT/2-1]/NFFT;
H = fftshift(abs(fft(h,NFFT)));
D = fftshift(abs(fft(d,NFFT)));

plot(f, D,'LineWidth',2);
title('frequency response of diff filter');
xlabel('frequency');
ylabel('amplitude');
```

```
saveas(gcf,'p6_freq_response_type3.png');

plot(h);
title('impulse response');
xlabel('samples');
ylabel('amplitude');
saveas(gcf,'p6_impulse_response_type3.png')

%type 4 filter
L = 22;
M = 21;
n = (0:M) - M/2;
h = cos(pi .* n)./(pi .* n) - sin(pi .* n)./(pi .* n.^2);

w = hamming(L).';
d = h.*w;

NFFT = 2^16;
f = [-NFFT/2:NFFT/2-1]/NFFT;
H = fftshift(abs(fft(h,NFFT)));
D = fftshift(abs(fft(d,NFFT)));

plot(f, D,'LineWidth',2);
title('frequency response of diff filter');
xlabel('frequency');
ylabel('amplitude');
saveas(gcf,'p6_freq_response_type4.png');

plot(h);
title('impulse response');
xlabel('samples');
ylabel('amplitude');
saveas(gcf,'p6_impulse_response_type4.png')
```